

# The Collected Mathematical Papers of Arthur Cayley

## Volume V

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Arthur Cayley

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### 349. On a Case of the Involution of Cubic Curves (*continued and concluded*)

[Continuation from p. 326. Arts. 36–87, treating the equation of the second tangent at a one-with-twofold centre, the alternative form of the nodal-tangent equation in terms of the node-coordinates, the triangle of critic centres, the harmonic-conic transformation, the three-centre conic, the transformation of the equation of the cubic, the harmonic conic and harmonoconic, and miscellaneous investigations.]

**36.** The equation of the other tangent at a one-with-twofold centre is

$$\beta^2\gamma x + \gamma^2\alpha y + \alpha^2\beta z = 0,$$

or, what is the same thing,

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 0.$$

This is in fact equivalent to

$$\begin{vmatrix} x & y & z \\ \alpha^2 & \beta^2 & \gamma^2 \\ \alpha^2(\beta - \gamma) & \beta^2(\gamma - \alpha) & \gamma^2(\alpha - \beta) \end{vmatrix} = 0;$$

for, developing the determinant, we find

$$x\beta^2\gamma^2(2\alpha - \beta - \gamma) + y\gamma^2\alpha^2(2\beta - \gamma - \alpha) + z\alpha^2\beta^2(2\gamma - \alpha - \beta) = 0,$$

or, what is the same thing (using  $\alpha + \beta + \gamma = 0$ ),

$$\beta^2\gamma^2 x + \gamma^2\alpha^2 y + \alpha^2\beta^2 z = 0,$$

hence the line

$$\alpha^2 \beta^2 \gamma^2 \left( \frac{x}{\alpha^2} + \frac{y}{\beta^2} + \frac{z}{\gamma^2} \right) = 0,$$

which is one of the tangents at the one-with-twofold centre, is also the line joining this point with the twofold centre.

**37.** The equation of the tangents at a critic centre or node may be obtained in a different form, involving, instead of the parameter  $\theta$ , the coordinates  $(x, y, z)$  of the node. We have

$$(\theta + \lambda)x = \theta(-x + y + z),$$

or, what is the same thing,

$$\lambda x = -\theta(x + y + z) + 2\theta x = -\theta(-x + y + z),$$

and similarly

$$\mu y = -\theta(x - y + z), \quad \nu z = -\theta(x + y - z);$$

thence also

$$\begin{aligned} (\theta + 4\lambda)x &= \theta(-x + 2y + 2z), \\ (\theta^2 + 4\mu\nu)yz &= \theta^2 \left\{ yz + \frac{1}{4}(x - y + z)(x + y - z) \right\} = \frac{1}{4}\theta^2(x^2 - y^2 - z^2 + 2yz). \end{aligned}$$

From which we obtain

$$a : b : c : f : g : h$$

proportional to

$$\begin{aligned} &2yz(-x + 2y + 2z), \quad 2zx(2x - y + z), \quad 2xy(2x + 2y - z), \\ &-x(-x^2 - y^2 - z^2 + 4yz), \quad -y(-x^2 + y^2 - z^2 + 4zx), \quad -z(-x^2 - y^2 + z^2 + 4xy), \end{aligned}$$

which are the required new forms.

**38.** We have

$$\begin{aligned} bc - f^2 &= 4xyz(-x + y + z)(2x + 2y - z) - x^2(-x^2 - y^2 - z^2 + 4yz)^2, \\ &= (x + y + z)^2(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy), \end{aligned}$$

which is = 0 if  $x + y + z = 0$ , or if  $x^2 - 2x(y + z) + (y - z)^2 = 0$ . In the former case, viz. if  $x + y + z = 0$ , we find  $a = b = c = f = g = h = -6xyz$ , and therefore

$$(a, b, c, f, g, h \mid x', y', z')^2 = -6xyz(x' + y' + z')^2,$$

but this corresponds merely to the value  $k = \infty$ , for which the cubic is

$$(x + y + z)^2(\lambda x + \mu y + \nu z) = 0,$$

which is not a proper cuspidal curve. In the latter case, or where

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0,$$

or, what is the same thing,  $\sqrt{x} + \sqrt{y} + \sqrt{z} = 0$ , we have a proper cuspidal curve.

*Article Nos. 39 to 43, relating to the Triangle of the Critic Centres.*

**39.** The equation

$$\frac{\lambda x}{\theta_1 + \lambda} + \frac{\mu y}{\theta_1 + \mu} + \frac{\nu z}{\theta_1 + \nu} = 0$$

is satisfied by substituting therein

$$x : y : z = \frac{1}{\theta_2 + \lambda} - \frac{1}{\theta_3 + \lambda} : \frac{1}{\theta_2 + \mu} - \frac{1}{\theta_3 + \mu} : \frac{1}{\theta_2 + \nu} - \frac{1}{\theta_3 + \nu};$$

in fact, for these values the equation becomes

$$\frac{\lambda}{(\theta_1 + \lambda)(\theta_2 + \lambda)} - \frac{\lambda}{(\theta_1 + \lambda)(\theta_3 + \lambda)} + \cdots = 0,$$

or as this may be written

$$\frac{1}{\theta_1 - \theta_3} \left( \frac{1}{\theta_1 + \lambda} - \frac{1}{\theta_3 + \lambda} \right) - \frac{1}{\theta_1 - \theta_2} \left( \frac{1}{\theta_1 + \lambda} - \frac{1}{\theta_2 + \lambda} \right) + \cdots = 0,$$

that is,  $1 - 1 = 0$ ; and similarly for the second set of values. Hence the equation in question is that of the line joining the critic centres corresponding to the roots  $\theta_2$  and  $\theta_3$ . Hence

$$\frac{\lambda x}{\theta_1 + \lambda} + \frac{\mu y}{\theta_1 + \mu} + \frac{\nu z}{\theta_1 + \nu} = 0,$$

$$\frac{\lambda x}{\theta_2 + \lambda} + \frac{\mu y}{\theta_2 + \mu} + \frac{\nu z}{\theta_2 + \nu} = 0,$$

$$\frac{\lambda x}{\theta_3 + \lambda} + \frac{\mu y}{\theta_3 + \mu} + \frac{\nu z}{\theta_3 + \nu} = 0,$$

are the equations of the sides of the triangle formed by the three critic centres.

**40.** It is to be remarked that the line

$$\frac{\lambda x}{\theta_1 + \lambda} + \frac{\mu y}{\theta_1 + \mu} + \frac{\nu z}{\theta_1 + \nu} = 0$$

is the polar of the critic centre  $(\frac{1}{\theta_1 + \lambda}, \frac{1}{\theta_1 + \mu}, \frac{1}{\theta_1 + \nu})$  in regard to the twofold-centre conic

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0;$$

in fact, forming the equation of the polar in question, this is

$$\left( -\frac{1}{\theta_1 + \lambda} + \frac{1}{\theta_1 + \mu} + \frac{1}{\theta_1 + \nu} \right) x + \cdots = 0,$$

but from the equation in  $\theta$ ,

$$\frac{1}{\theta_1 + \lambda} - \frac{1}{\theta_1 + \mu} - \frac{1}{\theta_1 + \nu} = \frac{2}{\theta_1} - \frac{2\lambda}{\theta_1(\theta_1 + \lambda)},$$

and the like for the coefficients of  $y$  and  $z$ ; this proves the theorem, and it thus appears that the critic centres are conjugate poles in regard to the twofold-centre conic.

*Article Nos. 41 to 50. Transformation of the Equation of the Nodal Tangents; the Three-Centre Conic.*

**41.** Writing as above,

$$l_1 = \theta_2 - \theta_3, \quad l_2 = \theta_3 - \theta_1, \quad l_3 = \theta_1 - \theta_2,$$

$$\mathfrak{S}_1 = (\theta_1 + \lambda)(\theta_1 + \mu)(\theta_1 + \nu), \quad \mathfrak{S}_2 = (\theta_2 + \lambda)(\theta_2 + \mu)(\theta_2 + \nu), \quad \mathfrak{S}_3 = (\theta_3 + \lambda)(\theta_3 + \mu)(\theta_3 + \nu),$$

I put for greater convenience

$$\begin{aligned} X &= -\frac{\theta_1}{l_1} \left( \frac{\lambda x}{\theta_1 + \lambda} + \frac{\mu y}{\theta_1 + \mu} + \frac{\nu z}{\theta_1 + \nu} \right), \\ Y &= -\frac{\theta_2}{l_2} \left( \frac{\lambda x}{\theta_2 + \lambda} + \frac{\mu y}{\theta_2 + \mu} + \frac{\nu z}{\theta_2 + \nu} \right), \\ Z &= -\frac{\theta_3}{l_3} \left( \frac{\lambda x}{\theta_3 + \lambda} + \frac{\mu y}{\theta_3 + \mu} + \frac{\nu z}{\theta_3 + \nu} \right), \end{aligned}$$

so that  $X = 0$ ,  $Y = 0$ ,  $Z = 0$  are the equations of the sides of the triangle formed by the critic centres.

**42.** Then  $X$ ,  $Y$ ,  $Z$  may, if we please, be considered as new coordinates replacing the original coordinates  $x$ ,  $y$ ,  $z$ ; the relation between the two sets being given by the equations last written down; the values of  $x$ ,  $y$ ,  $z$  in terms of  $X$ ,  $Y$ ,  $Z$  are given by the converse system

$$x = \frac{X}{\theta_1 + \lambda} + \frac{Y}{\theta_2 + \lambda} + \frac{Z}{\theta_3 + \lambda}, \quad y = \frac{X}{\theta_1 + \mu} + \frac{Y}{\theta_2 + \mu} + \frac{Z}{\theta_3 + \mu}, \quad z = \frac{X}{\theta_1 + \nu} + \frac{Y}{\theta_2 + \nu} + \frac{Z}{\theta_3 + \nu}.$$

**43.** To exhibit the identity of the two systems, I start from the last-mentioned one; this gives, on substitution,

$$\frac{\lambda x}{\theta_1 + \lambda} = X \left( \frac{1}{\theta_1 + \lambda} - \frac{1}{\theta_2 + \lambda} - \frac{1}{\theta_3 + \lambda} \right) + \dots,$$

where the coefficient of  $X$  reduces, after a short calculation, to

$$\frac{(\mu - \nu)(\nu - \lambda)(\lambda - \mu)(\theta_1 - \theta_2)(\theta_2 - \theta_3)(\theta_3 - \theta_1)}{(\theta_1 + \lambda)(\theta_1 + \mu)(\theta_1 + \nu)(\theta_2 + \lambda)(\theta_2 + \mu)(\theta_2 + \nu)(\theta_3 + \lambda)(\theta_3 + \mu)(\theta_3 + \nu)},$$

or, what is the same thing,

$$\frac{-(\mu - \nu)(\nu - \lambda)(\lambda - \mu) l_1 l_2 l_3}{\mathfrak{S}_1 \mathfrak{S}_2 \mathfrak{S}_3}.$$

The first side is a linear function of  $x$ ,  $y$ ,  $z$  which vanishes for

$$x : y : z = \frac{1}{\theta_2 + \lambda} : \frac{1}{\theta_2 + \mu} : \frac{1}{\theta_2 + \nu},$$

and for

$$x : y : z = \frac{1}{\theta_3 + \lambda} : \frac{1}{\theta_3 + \mu} : \frac{1}{\theta_3 + \nu},$$

and hence it is of the form

$$K \left( \frac{\lambda x}{\theta_1 + \lambda} + \frac{\mu y}{\theta_1 + \mu} + \frac{\nu z}{\theta_1 + \nu} \right),$$

and by comparing coefficients of  $x$  we have

$$\frac{K\lambda}{\theta_1 + \lambda} = \frac{1}{(\theta_2 + \mu)(\theta_2 + \nu)} - \frac{1}{(\theta_3 + \mu)(\theta_3 + \nu)} = \frac{(\theta_3 - \theta_2)(\theta_2 + \theta_3 + \mu + \nu)}{(\theta_2 + \mu)(\theta_2 + \nu)(\theta_3 + \mu)(\theta_3 + \nu)};$$

that is

$$K = \frac{-l_1(\mu - \nu)(\theta_1 + \lambda)}{(\theta_2 + \mu)(\theta_2 + \nu)(\theta_3 + \mu)(\theta_3 + \nu)\lambda},$$

and it is easy to see that

$$K = \frac{-\theta_1}{l_1},$$

so that, using

$$(\theta_1 + \lambda)(\theta_2 + \lambda)(\theta_3 + \lambda) = -\lambda(\nu - \lambda)(\lambda - \mu),$$

the equation becomes

$$X = -\frac{\theta_1}{l_1} \left( \frac{\lambda x}{\theta_1 + \lambda} + \frac{\mu y}{\theta_1 + \mu} + \frac{\nu z}{\theta_1 + \nu} \right),$$

which is right; and similarly for the values of  $Y$  and  $Z$ .

**44.** The equation of the tangents at the node corresponding to the root  $\theta_1$  is

$$\left( \theta_1 + 4\lambda, \theta_1 + 4\mu, \theta_1 + 4\nu, -\theta_1 - \frac{2\mu\nu}{\theta_1 + \lambda}, -\theta_1 - \frac{2\nu\lambda}{\theta_1 + \mu}, -\theta_1 - \frac{2\lambda\mu}{\theta_1 + \nu} \right) (x, y, z)^2 = 0;$$

and substituting for  $x, y, z$  their values in terms of  $X, Y, Z$ , it appears in the first place that the coefficients of  $X^2, XY, XZ, YZ$  all of them vanish.

**45.** In fact,

$$\text{coeff. } X^2 = (\theta_1 + 4\lambda) \frac{1}{(\theta_1 + \lambda)^2} + \dots - 2\left(\theta_1 + \frac{2\mu\nu}{\theta_1 + \lambda}\right) \frac{1}{(\theta_1 + \mu)(\theta_1 + \nu)} - \dots.$$

The first term is

$$\sum \frac{\theta_1 + 4\lambda}{(\theta_1 + \lambda)^2} = \sum \frac{1}{\theta_1 + \lambda} + 3 \sum \frac{\lambda}{(\theta_1 + \lambda)^2} = \frac{2}{\theta_1} + \frac{3}{\theta_1^2} \{3\theta_1^2 - (\mu\nu + \nu\lambda + \lambda\mu)\},$$

where the value of  $\sum \frac{\lambda}{(\theta_1 + \lambda)^2}$  is most readily found from the identical equation

$$\frac{\lambda}{\theta_1 + \lambda} + \frac{\mu}{\theta_1 + \mu} + \frac{\nu}{\theta_1 + \nu} = \frac{\theta^3 - \theta(\mu\nu + \nu\lambda + \lambda\mu) - \lambda\mu\nu}{(\theta + \lambda)(\theta + \mu)(\theta + \nu)}$$

by differentiating and then writing  $\theta = \theta_1$ . The second term is

$$-\frac{2}{\theta_1^2} \{3\theta_1^2 + (\lambda + \mu + \nu)\theta_1 + 2(\mu\nu + \nu\lambda + \lambda\mu) + \frac{2\lambda\mu\nu}{\theta_1}\}.$$

Whole is  $\frac{2}{\theta_1^2}$  multiplied into

$$\begin{aligned} & \theta_1^2 + 3\{3\theta_1^2 - (\mu\nu + \nu\lambda + \lambda\mu)\} - 2\{3\theta_1^2 + (\lambda + \mu + \nu)\theta_1 + 2(\mu\nu + \nu\lambda + \lambda\mu) + \frac{2\lambda\mu\nu}{\theta_1}\} \\ &= \frac{2}{\theta_1^2} \{\theta_1^3 - (\mu\nu + \nu\lambda + \lambda\mu)\theta_1 - 2\lambda\mu\nu\}, \end{aligned}$$

which is = 0.

**46.** We have next

$$\text{coeff. } XY = (\theta_1 + 4\lambda, \dots, -\theta_1 - \frac{2\mu\nu}{\theta_1 + \lambda}, \dots) \left( \frac{1}{\theta_1 + \lambda}, \frac{1}{\theta_1 + \mu}, \frac{1}{\theta_1 + \nu} \right) \left( \frac{1}{\theta_2 + \lambda}, \frac{1}{\theta_2 + \mu}, \frac{1}{\theta_2 + \nu} \right).$$

This expands as

$$\sum \frac{\theta_1 + 4\lambda}{(\theta_1 + \lambda)(\theta_2 + \lambda)} - \sum (\theta_1 + \frac{2\mu\nu}{\theta_1 + \lambda}) \left\{ \frac{1}{(\theta_1 + \mu)(\theta_2 + \nu)} + \frac{1}{(\theta_1 + \nu)(\theta_2 + \mu)} \right\}.$$

First term reduces to

$$\sum \frac{\theta_1 + 4\lambda}{(\theta_1 + \lambda)(\theta_2 + \lambda)} = \sum \frac{1}{\theta_2 - \theta_1} \left( \frac{1}{\theta_1 + \lambda} - \frac{1}{\theta_2 + \lambda} \right) + 3 \sum \frac{\lambda}{(\theta_1 + \lambda)(\theta_2 + \lambda)},$$

and after using

$$\sum \frac{\lambda}{(\theta_1 + \lambda)(\theta_2 + \lambda)} = \frac{\lambda\mu\nu}{\theta_1\theta_2} \cdot \frac{1}{\theta_2 - \theta_1} \left( \frac{2}{\theta_1} - \frac{2}{\theta_2} \right) = \frac{2}{\theta_1\theta_2},$$

one finds that the first term equals  $\frac{2}{\theta_1\theta_2}$ .

The second term, written out at full length and collecting the terms which contain  $\frac{1}{\theta_1+\lambda}$ , becomes

$$-\frac{1}{\theta_1\theta_2} \sum \frac{2\mu\nu}{(\theta_1+\lambda)} - \frac{1}{\theta_1\theta_2} \sum \frac{1}{\theta_1-\theta_2} \left( \frac{2\lambda^2}{\theta_1+\lambda} - \frac{2\lambda^2}{\theta_2+\lambda} \right).$$

Observing that

$$\frac{2\mu\nu}{\theta_1+\lambda} = 3\theta_1 + (\lambda + \mu + \nu) - 6(\theta_1 + \lambda) - 2\lambda = -3\theta_1 + \lambda + \mu + \nu,$$

together with the analogous expression for  $\frac{2\nu\lambda}{\theta_2+\lambda}$ , the second part reduces to  $-\frac{3}{\theta_1\theta_2}$ .

47. Hence the whole second term is  $-\frac{2}{\theta_1\theta_2}$ , and combining the two terms we have

$$\text{coeff. } XY = \frac{2}{\theta_1\theta_2} - \frac{2}{\theta_1\theta_2} = 0.$$

In the same manner precisely it appears that

$$\text{coeff. } XZ = 0.$$

48. Next,

$$\text{coeff. } YZ = (\theta_1 + 4\lambda, \dots, -\theta_1 - \frac{2\mu\nu}{\theta_1+\lambda}, \dots) \left( \frac{1}{\theta_2+\lambda}, \frac{1}{\theta_2+\mu}, \frac{1}{\theta_2+\nu} \right) \left( \frac{1}{\theta_3+\lambda}, \frac{1}{\theta_3+\mu}, \frac{1}{\theta_3+\nu} \right).$$

The first term is

$$\sum \frac{\theta_1 + 4\lambda}{(\theta_2 + \lambda)(\theta_3 + \lambda)} = \sum \frac{1}{\theta_3 - \theta_2} \left( \frac{1}{\theta_2 + \lambda} - \frac{1}{\theta_3 + \lambda} \right) + 3 \sum \frac{\lambda}{(\theta_2 + \lambda)(\theta_3 + \lambda)},$$

which reduces to

$$\frac{2}{\theta_2\theta_3} \cdot \frac{\theta_1 l_1}{\theta_2 - \theta_3} \left( \frac{1}{\theta_2} - \frac{1}{\theta_3} \right) - \frac{l_1}{\theta_2\theta_3} = -\frac{l_1}{\theta_2\theta_3}.$$

The second term, written out at full length and rearranged, is easily seen to be

$$-\frac{l_1}{\theta_2\theta_3} \left\{ \sum \frac{2\mu\nu}{\theta_1 + \lambda} \cdot \frac{1}{\theta_2 - \theta_3} \left( \frac{1}{\theta_2 + \mu} + \frac{1}{\theta_3 + \mu} \right) - \dots \right\},$$

so that the whole second term reduces to  $-\frac{l_1}{\theta_2\theta_3}$ ; whence combining the two we have

$$\text{coeff. } YZ = \frac{l_1}{\theta_2\theta_3} - \frac{l_1}{\theta_2\theta_3} = 0.$$

49. We have now to find the coefficients of  $Y^2$  and  $Z^2$ .

$$\text{coeff. } Y^2 = (\theta_1 + 4\lambda, \dots, -\theta_1 - \frac{2\mu\nu}{\theta_1+\lambda}, \dots) \left( \frac{1}{\theta_2+\lambda}, \frac{1}{\theta_2+\mu}, \frac{1}{\theta_2+\nu} \right)^2,$$

and observing that the terms of this expression differ from those of  $\text{coeff. } X^2$  only by having  $\theta_2$  in place of  $\theta_1$ , so that the corresponding part vanishes, the only residue arises from the coefficients involving  $\theta_1$  explicitly. After collection,

$$\text{coeff. } Y^2 = (\theta_1 - \theta_2) \left\{ \sum \frac{1}{(\theta_2 + \lambda)^2} - 2 \sum \frac{1}{(\theta_2 + \mu)(\theta_2 + \nu)} \right\} + \frac{2}{\theta_1} (2(\lambda + \mu + \nu) + \frac{2\mu\nu}{\theta_1 + \lambda} + \frac{2\nu\lambda}{\theta_1 + \mu} + \frac{2\lambda\mu}{\theta_1 + \nu}),$$

and substituting for  $\theta_1 - \theta_2$  its value  $l_3$ , one obtains

$$\text{coeff. } Y^2 = \frac{l_3}{\theta_2^2} \{ -3\theta_2^2 + (\mu\nu + \nu\lambda + \lambda\mu)(\theta_2 + 2\theta_1) + 6\lambda\mu\nu \}.$$

But  $\theta_1 + \theta_2 + \theta_3 = 0$ , whence  $\theta_2 + 2\theta_1 = \theta_1 - \theta_3 = l_2$ ; and  $\mu\nu + \nu\lambda + \lambda\mu = -(\theta_1\theta_2 + \theta_2\theta_3 + \theta_3\theta_1)$ ,  $\lambda\mu\nu = -\theta_1\theta_2\theta_3$ ; so

$$\text{coeff. } Y^2 = \frac{l_3}{\theta_2^2} \{-3\theta_2^2 - (\theta_1\theta_2 + \theta_2\theta_3 + \theta_3\theta_1)l_2 - 6\theta_1\theta_2\theta_3\},$$

that is

$$\text{coeff. } Y^2 = -\frac{l_3 l_2}{\theta_2},$$

and, by merely interchanging  $\theta_2$  and  $\theta_3$ ,

$$\text{coeff. } Z^2 = -\frac{l_2 l_3}{\theta_3}.$$

**50.** Hence the equation of the tangents is

$$-\frac{l_3 l_2}{\theta_2} Y^2 - \frac{l_2 l_3}{\theta_3} Z^2 = 0,$$

or, what is the same thing,

$$\frac{Y^2}{\theta_2} + \frac{Z^2}{\theta_3} = 0,$$

or putting

$$A = \frac{l_1^2}{\theta_1}, \quad B = \frac{l_2^2}{\theta_2}, \quad C = \frac{l_3^2}{\theta_3},$$

the equation of the tangents at the node corresponding to  $\theta_1$  is  $BY^2 + CZ^2 = 0$ . Hence the equations of the tangents at the three nodes respectively are

$$BY^2 + CZ^2 = 0,$$

$$AX^2 + CZ^2 = 0,$$

$$AX^2 + BY^2 = 0;$$

that is, the nodes or critic centres are conjugate poles in regard to a conic

$$AX^2 + BY^2 + CZ^2 = 0,$$

which is the *three-centre conic*; and the tangents at each node are the tangents from such node to the conic in question.

*Article Nos. 51 and 52. Special Case of the Three-Centre Conic.*

**51.** Write for a moment

$$X' = \frac{\lambda x}{\theta_1 + \lambda} + \frac{\mu y}{\theta_1 + \mu} + \frac{\nu z}{\theta_1 + \nu},$$

$$Y' = \frac{\lambda x}{\theta_2 + \lambda} + \frac{\mu y}{\theta_2 + \mu} + \frac{\nu z}{\theta_2 + \nu},$$

$$Z' = \frac{\lambda x}{\theta_3 + \lambda} + \frac{\mu y}{\theta_3 + \mu} + \frac{\nu z}{\theta_3 + \nu},$$

so that  $X = -\frac{\theta_1}{l_1} X'$ , etc.; the equation of the three-centre conic in terms of  $(X', Y', Z')$  is

$$\frac{l_1^2}{\theta_1} \frac{\theta_1^2}{l_1^2} X'^2 + \frac{l_2^2}{\theta_2} \frac{\theta_2^2}{l_2^2} Y'^2 + \frac{l_3^2}{\theta_3} \frac{\theta_3^2}{l_3^2} Z'^2 = 0,$$

say

$$A'X'^2 + B'Y'^2 + C'Z'^2 = 0.$$

When  $\theta_3 = \infty$ , we have  $C' = \infty$ ,  $A' = -B'$ ,  $X' = Y'$ ; writing the equation in the form

$$(A' + B')X'^2 + B'(Y'^2 - X'^2) + C'Z'^2 = 0,$$

and observing that in the limit  $Y'^2 - X'^2 = 2X'(Y' - X')$ , we see that the equation will assume the form

$$(\theta_1\theta_2 - \theta_1^2)X'\Sigma - \frac{l_1}{\theta_1 + 2\theta_1}\theta_2Z'^2 = 0,$$

where  $\Sigma$  is a finite function;  $X' = 0$  is the line joining the twofold centre and the one-with-twofold centre,  $\Sigma = 0$  is the other tangent at the one-with-twofold centre,  $Z' = 0$  the tangent at the twofold centre or cusp; the form  $X'\Sigma + Z'^2 = 0$  shows that the three-centre conic reduces itself to a pair of points, viz. the twofold centre or cusp, and the point where the tangent at the cusp is met by the other tangent (that is the tangent not passing through the cusp) at the one-with-twofold centre.

**52.** To verify the value of  $\Sigma$  I proceed as follows:

$$A' + B' = \frac{1}{\theta_1 - \theta_2} \{ \theta_1(\theta_2 - \theta_3) + \theta_2(\theta_3 - \theta_1) \} = (\theta_1 - \theta_2)(\lambda + \mu + \nu + 3\theta_1);$$

$$\theta_1^3 - \theta_2^3 + (\lambda + \mu + \nu)(\theta_1^2 - \theta_2^2) + (\mu\nu + \nu\lambda + \lambda\mu)(\theta_1 - \theta_2) = (\theta_1 - \theta_2) \{ 6\theta_1^2 + 2(\lambda + \mu + \nu)\theta_1 \}.$$

Moreover

$$Y'^2 - X'^2 = 2X' \left\{ \lambda x \left( \frac{1}{\theta_1 + \lambda} \cdot \frac{1}{\theta_2 + \lambda} \right) + \mu y \left( \frac{1}{\theta_1 + \mu} \cdot \frac{1}{\theta_2 + \mu} \right) + \nu z \left( \frac{1}{\theta_1 + \nu} \cdot \frac{1}{\theta_2 + \nu} \right) \right\},$$

and hence

$$B'(Y'^2 - X'^2) = -\frac{2\theta_1}{l_1} X' \Sigma',$$

in which we have only now to substitute  $(\lambda, \mu, \nu) = (\alpha^{-2}, \beta^{-2}, \gamma^{-2})$  and  $\theta_1 = -\frac{1}{\alpha\beta\gamma}$ . We have

$$\theta_1 + \lambda = \frac{\beta\gamma}{\alpha^2\beta^2\gamma^2} k_1, \quad \theta_1 + \mu = \frac{\gamma\alpha}{\alpha^2\beta^2\gamma^2} k_2, \quad \theta_1 + \nu = \frac{\alpha\beta}{\alpha^2\beta^2\gamma^2} k_3,$$

where  $k = -3(\beta\gamma - \alpha^2) = -3(\beta\gamma + \gamma\alpha + \alpha\beta)$ ; and then observing that the equation  $\Sigma = 0$  becomes

$$2(\beta\gamma + \gamma\alpha + \alpha\beta) \left( \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} \right) + 3(\alpha x + \beta y + \gamma z) = 0,$$

or, what is the same thing,

$$(2\beta\gamma + \alpha^2) \frac{x}{\alpha} + (2\gamma\alpha + \beta^2) \frac{y}{\beta} + (2\alpha\beta + \gamma^2) \frac{z}{\gamma} = 0,$$

which agrees with a former result.

*Article Nos. 53 to 55. Transformation of the Equation of the Cubic.*

**53.** Let it be required to express the cubic

$$xyz + k(x + y + z)^2(\lambda x + \mu y + \nu z) = 0$$

in terms of the coordinates  $X, Y, Z$ . We have

$$\lambda x + \mu y + \nu z = X + Y + Z,$$



and the equation therefore is

$$\Pi\left(\frac{X}{\theta_1 + \lambda} + \frac{Y}{\theta_2 + \lambda} + \frac{Z}{\theta_3 + \lambda}\right) + k\left(\frac{X}{\theta_1} + \frac{Y}{\theta_2} + \frac{Z}{\theta_3}\right)^2 (X + Y + Z) = 0,$$

where  $\Pi$  denotes the product of the three factors obtained by writing  $\lambda, \mu, \nu$  successively in the place of  $\lambda$ .

For one of the nodal cubics we have

$$k = -\frac{1}{4}\theta_1,$$

and the equation multiplied by  $\theta_1$  is

$$\theta_1 \Pi(\dots) - \frac{1}{4}\theta_1^2\left(\frac{X}{\theta_1} + \frac{Y}{\theta_2} + \frac{Z}{\theta_3}\right)^2 (X + Y + Z) = 0,$$

which it is clear *à priori* must be of the form

$$(\dots + X)^2(\dots)(X + Y + Z) = 0,$$

and there is in fact no difficulty in verifying that the coefficients of  $X^3, X^2Y, X^2Z, XYZ$  all of them vanish. To find  $K$ , comparing the coefficients of  $XY^2$  we have

$$K = 2(\theta_1 + \lambda)(\theta_2 + \mu)(\theta_2 + \nu) - \frac{1}{4}\theta_1(\theta_1 + 2\theta_2)\theta_2;$$

that is

$$K = \frac{1}{4}(\theta_1 - \theta_2)l_2^2 + \frac{1}{4}\Sigma(\theta_1 + \theta_3)\theta_2,$$

and similarly

$$K = \frac{1}{4}\{\theta_1^2\theta_2^2 - 6\theta_1\theta_2(\mu\nu + \nu\lambda + \lambda\mu) - (\theta_1 + \theta_2)^2\theta_3^2\} = \frac{1}{4}\{\theta_1^2\theta_2^2 + 6\theta_1\theta_2(\theta_1\theta_2 + \theta_2\theta_3 + \theta_3\theta_1) - (\theta_1 + \theta_2)^2\theta_3^2\},$$

so that we have

$$K = \frac{1}{4}\theta_3^2l_3^2,$$

that is

$$K = \frac{1}{4}\theta_3l_3^2,$$

and the equation of the nodal cubic is

$$-\frac{1}{4}\Pi X\left(\frac{Y^2}{\theta_2^2l_2^2} + \frac{Z^2}{\theta_3^2l_3^2}\right) + \frac{1}{4}\theta_1\Pi\left(\frac{Y}{\theta_2l_2} - \frac{Z}{\theta_3l_3}\right)^2 \cdot \Sigma = 0.$$

**54.** To complete the reduction we have

$$\text{coeff. } Y^2 = \frac{1}{4} \cdot \frac{\theta_1\theta_3^2l_3^2}{\theta_2^2} \cdot (\theta_1\theta_2^2l_2^2/(\theta_2^2l_2^2)),$$

$$\text{coeff. } YZ = \frac{\theta_1}{2} \cdot \frac{l_3l_2}{\theta_2\theta_3} \cdot (\theta_2\theta_3^2l_3^2 - \theta_2^2\theta_3l_2^2),$$

so that substituting for  $\sqrt{\theta_1 - \theta_2\theta_3}$  its value  $-\frac{1}{2}\theta_1l_1^2$ , the terms in  $Y^2$  and  $YZ$  are

...

and in like manner the terms in  $YZ^2$  and  $Z^3$  are

...

**55.** So that the terms in  $(Y, Z)^3$  are

$$-\frac{1}{4}\theta_1^2 l_1^2 \left\{ \frac{Y^3}{\theta_2 l_2} \left( \frac{1}{\theta_2} - \frac{1}{\theta_3} l_2^2 \right) - \frac{Z^3}{\theta_3 l_3} \left( \frac{1}{\theta_3} - \frac{1}{\theta_2} l_3^2 \right) \right\},$$

and the equation, omitting the factor  $-\frac{1}{4}\theta_1^2 l_1^2$ , is

$$\frac{Y}{\theta_2^2 l_2^3} - \frac{Z}{\theta_3^2 l_3^3} = 0.$$

But the term in [...] is

$$\frac{Y}{\theta_2^2 l_2^3} - \frac{Z}{\theta_3^2 l_3^3} = \frac{1}{l_1^2 \theta_2^2 \theta_3^2} \left( \frac{l_3 Y}{\theta_2 l_2} - \frac{l_2 Z}{\theta_3 l_3} \right),$$

and the equation of the nodal cubic is finally

$$\frac{Y}{\theta_2^2 l_2^3} - \frac{Z}{\theta_3^2 l_3^3} = 0.$$

The lines  $Y = 0$ ,  $Z = 0$ ,  $\frac{Y}{\theta_2^2 l_2^3} - \frac{Z}{\theta_3^2 l_3^3} = 0$  each pass through the node and meet the cubic in a third point; the three points of intersection lie in the line  $l_1 X + \theta_1(Y - Z) = 0$ .

*Article Nos. 56 to 66. The Cubic Locus, Harmonoconics and Harmonic Conic.*

**56.** Suppose that the line  $\lambda x + \mu y + \nu z = 0$  passes through a given point  $(a, b, c)$ , then we have

$$\lambda a + \mu b + \nu c = 0;$$

and observing that  $\theta + \lambda$ ,  $\theta + \mu$ ,  $\theta + \nu$ ,  $\theta$  are proportional to

$$\frac{1}{x}, \quad \frac{1}{y}, \quad \frac{1}{z}, \quad \frac{2}{x+y+z}, \quad \text{respectively,}$$

we find

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = \frac{2(a+b+c)}{x+y+z},$$

the equation of a cubic curve, the locus of the critic centres corresponding to the several lines  $\lambda x + \mu y + \nu z = 0$  which pass through the point  $(a, b, c)$ . The cubic curve passes, it is clear, through the six points which are the angles of the quadrilateral

$$x = 0, \quad y = 0, \quad z = 0, \quad x + y + z = 0.$$

**57.** If we take for  $(a, b, c)$  the coordinates of the point of intersection of the line  $\lambda x + \mu y + \nu z = 0$  with any one of the lines  $x = 0$ ,  $y = 0$ ,  $z = 0$ ,  $x + y + z = 0$ , then in each case the cubic breaks up into the same line and a conic, viz. we have the conics

$$\frac{\nu}{y} - \frac{\mu}{z} - \frac{2(\nu - \mu)}{x + y + z} = 0,$$

$$\frac{\lambda}{z} - \frac{\nu}{x} - \frac{2(\lambda - \nu)}{x + y + z} = 0,$$

$$\frac{\mu}{x} - \frac{\lambda}{y} - \frac{2(\mu - \lambda)}{x + y + z} = 0,$$

$$\frac{\lambda}{x} + \frac{\mu}{y} + \frac{\nu}{z} = 0,$$

and it is to be noticed that in each case the critic centres all of them lie on the conic. In fact, since the point  $(0, \nu, -\mu)$  is an arbitrary point on the line  $x = 0$ , a line  $\lambda x + \mu y + \nu z = 0$  passing

through the point in question is an absolutely arbitrary line, and the corresponding critic centres therefore do not lie on the line  $x = 0$ ; that is, they lie on the conic

$$\frac{\nu}{y} - \frac{\mu}{z} - \frac{2(\nu - \mu)}{x + y + z} = 0,$$

and it may also be remarked that the elimination of  $\lambda, \theta$  from the system

$$\theta + \mu : \theta + \nu : \theta = \frac{1}{y} : \frac{1}{z} : \frac{2}{x+y+z},$$

gives the last-mentioned equation, unencumbered by the factor  $x = 0$ .

We have thus four conics, each of them passing through the three critic centres which correspond to the line  $\lambda x + \mu y + \nu z = 0$ ; as to the signification of the first three of these conics, I remark as follows.

**58.** The ‘harmonic’ of a point  $A$  as to the line  $T$  in respect of the conic  $\Omega$  may be defined as follows; viz. considering the pencil of lines through  $A$ , the locus of the fourth harmonic of the point in which a line of the pencil meets  $T$ , in regard to the two points in which the same line meets the conic  $\Omega$ , is a conic which is the harmonic in question. (In particular, if the line  $T$  pass through the point  $A$  the harmonic breaks up into the line  $T$  and into the polar of  $A$ .) The conic  $\Omega$  may of course be a pair of lines.

Consider any three lines  $x, y, z$ , a line  $S$ , and the line  $T$ ; then the harmonics being all as to the same line  $T$ , we have the theorem

- Harmonic of intersection of  $x, S$  in regard to pair of lines  $y, z$ ,
- Ditto of  $y, S$  in regard to pair of lines  $z, x$ ,
- Ditto of  $z, S$  in regard to pair of lines  $x, y$ ,

all pass through the same three points.

And taking  $x = 0, y = 0, z = 0$  for the equations of the lines  $x, y, z$ ;  $\lambda x + \mu y + \nu z = 0$  for the equation of the line  $S$ ; and  $x + y + z = 0$  for the equation of the line  $T$ , the harmonics just spoken of are the above-mentioned three conics respectively.

**59.** In fact, considering the harmonic of intersection of  $x, S$  in regard to the pair  $y, z$ ; and taking  $x', y', z'$  as the coordinates of a point  $P$  of the harmonic, then the equation of the line  $AP$  is

$$\det \begin{pmatrix} x & y & z \\ x' & y' & z' \\ 0 & \nu & -\mu \end{pmatrix} = 0,$$

that is

$$x(\mu y' + \nu z') - x'(\mu y + \nu z) = 0,$$

and at the point of intersection with the line  $T$  or  $x + y + z = 0$ , we have

$$(y + z)(\mu y' + \nu z') + x'(\mu y + \nu z) = 0,$$

or, what is the same thing,

$$y(\mu x' + \mu y' + \nu z') + z(\nu x' + \mu y' + \nu z') = 0,$$

which is the line through the last-mentioned point and the point  $(y = 0, z = 0)$ .

The line from the point  $A$  to the point  $(y = 0, z = 0)$  is

$$y z' - z y' = 0.$$

**60.** By the definition of the harmoconic, the last-mentioned two lines are harmonics in regard to the lines  $y = 0, z = 0$ ; that is, we have for the equation of the harmoconic in question

$$-y'(\mu x' + \mu y' + \nu z') + z'(\nu x' + \mu y' + \nu z') = 0,$$

this equation may also be written

$$(yz' - zy')(\mu x' + y' + z') - 2(\nu - \mu)y'z' = 0,$$

or, what is the same thing,

$$\frac{\nu}{y} - \frac{\mu}{z} - \frac{2(\nu - \mu)}{x + y + z} = 0,$$

hence writing  $x, y, z$  in place of  $x', y', z'$ , we see that this harmoconic is in fact the first of the above-mentioned three conics.

**61.** The fourth conic through the critic centres is the conic

$$\frac{\lambda}{x} + \frac{\mu}{y} + \frac{\nu}{z} = 0,$$

which it will be observed passes through the vertices of the triangle  $x = 0, y = 0, z = 0$ , and also through the point  $(1, 1, 1)$ , which is the harmonic of the line  $x + y + z = 0$  in regard to the triangle: I call it the 'harmonic conic.' Representing the equation by

$$2fyz + 2gzx + 2hxy = 0,$$

we have  $f = \mu - \nu, g = \nu - \lambda, h = \lambda - \mu$ , and therefore  $f + g + h = 0$ .

**62.** It is easy to show that the coordinates of the pole of the line  $x + y + z = 0$  in regard to the harmonic conic are  $x : y : z = f^2 : g^2 : h^2$ ; these values satisfy the condition  $\sqrt{x} + \sqrt{y} + \sqrt{z} = 0$ , that is, the pole in question lies on the twofold-centre conic.

**63.** The equation of the tangents to the harmonic conic at its intersection with the line  $x + y + z = 0$  (which tangents meet of course in the last-mentioned pole, that is in a point of the twofold-centre conic) is found to be

$$2fgh(x + y + z)^2 + \Omega(2fyz + 2gzx + 2hxy) = 0;$$

if for shortness

$$\Omega = f^2 + g^2 + h^2 - 2gh - 2hf - 2fg,$$

or, what is the same thing,

$$\Omega = -4(gh + hf + fg) = 2(f^2 + g^2 + h^2).$$

**64.** We have identically

$$\begin{aligned} & -6fgh(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy) + (hz)(ghx + hfy + fgz) \\ & = 2fgh(x + y + z)^2 + \Omega(2fyz + 2gzx + 2hxy) - 8(fx + gy + hz)(ghx + hfy + fgz), \end{aligned}$$

so that the tangents in question meet the twofold-centre conic

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0$$

at its intersections with the lines  $fx + gy + hz = 0$  and  $ghx + hfy + fgz = 0$ : the latter of these is in fact the tangent of the conic at the point  $(f^2, g^2, h^2)$  of intersection of the two tangents. Hence

the two tangents meet at the point  $(f^2, g^2, h^2)$  of the twofold-centre conic and they besides meet the conic at its points of intersection with the line  $fx + gy + hz = 0$ .

**65.** The line  $\lambda x + \mu y + \nu z = 0$  may be expressed in the form,

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} + \frac{1}{3}(x + y + z) = 0,$$

(where, *ut supra*,  $\alpha + \beta + \gamma = 0$ ). The corresponding values of  $f, g, h$  are

$$f = \frac{1}{\beta} - \frac{1}{\gamma}, \quad g = \frac{1}{\gamma} - \frac{1}{\alpha}, \quad h = \frac{1}{\alpha} - \frac{1}{\beta},$$

or, what is the same thing,

$$f : g : h = \alpha^2(\beta - \gamma) : \beta^2(\gamma - \alpha) : \gamma^2(\alpha - \beta),$$

or, again,

$$f : g : h = \alpha(\beta^2 - \gamma^2) : \beta(\gamma^2 - \alpha^2) : \gamma(\alpha^2 - \beta^2).$$

The equation  $fx + gy + hz = 0$  may be written

$$\det \begin{pmatrix} x & y & z \\ \frac{1}{\alpha^2} & \frac{1}{\beta^2} & \frac{1}{\gamma^2} \\ 1 & 1 & 1 \end{pmatrix} = 0,$$

that is, the line in question is the line joining the harmonic point  $(1, 1, 1)$  with the inverse of the point  $(\alpha^2, \beta^2, \gamma^2)$ , which is (*ante*, No. 27) the point of contact of the satellite line with the envelope.

**66.** The harmonic conic passes through the vertices of the triangle  $x = 0, y = 0, z = 0$ , through the harmonic point  $(1, 1, 1)$ , and through the critic centres. Hence if one of the critic centres be given, the harmonic conic passes through five given points and is thus completely determined. But a critic centre being given, the line joining the other two critic centres is the polar of the given centre in regard to the twofold-centre conic (*ante*, No. 40), and it is thus completely determined; and the other two critic centres are of course the intersections of this line with the harmonic conic.

*Article Nos. 67 to 87. Miscellaneous Investigations.*

**67.** I demonstrate by means of the last-mentioned formulae a theorem already in effect demonstrated by the investigation which led to the three-centre conic, viz. that the tangents at a node or critic centre, and the lines drawn to the other two critic centres, form a harmonic pencil.

In fact the tangents at the node or critic centre are given by the equation

$$(\theta + 4\lambda, \dots, -\theta - \frac{2\mu\nu}{\theta + \lambda}, \dots)(x, y, z)^2 = 0;$$

the other two critic centres are given as the intersection of the line

$$\frac{\lambda x}{\theta + \lambda} + \frac{\mu y}{\theta + \mu} + \frac{\nu z}{\theta + \nu} = 0$$

with the conic

$$\frac{\lambda}{x} + \frac{\mu}{y} + \frac{\nu}{z} = 0;$$

the theorem will be true if the pair of tangents and the last-mentioned conic are cut harmonically by the last-mentioned line.

Now in general the condition in order that the line  $\xi x + \eta y + \zeta z = 0$  may cut harmonically the conics  $(a, b, c, f, g, h \mid x, y, z)^2$  and  $(a', b', c', f', g', h')(x, y, z)^2 = 0$  is

$$(bc' + b'c - 2ff', \dots, gh' + g'h - af' - a'f, \dots \mid \xi, \eta, \zeta)^2 = 0,$$

and if  $a' = b' = c' = 0$ , then the condition is

$$(-2ff', \dots, gh' + g'h - af', \dots \mid \xi, \eta, \zeta)^2 = 0.$$

**68.** In the present case the equations of the two conics may be written

$$(\theta + 4\lambda, \dots, -\theta - \frac{2\mu\nu}{\theta + \lambda}, \dots)(x, y, z)^2 = 0,$$

$$(0, \dots, \mu - \nu, \dots)(x, y, z)^2 = 0,$$

and we have

$$\begin{aligned} -2ff' &= -2(\mu - \nu)(\theta + \frac{2\mu\nu}{\theta + \lambda}), \\ gh' + g'h - af' &= -(\theta + \frac{2\nu\lambda}{\theta + \mu})(\lambda - \mu) - (\theta + \frac{2\lambda\mu}{\theta + \nu})(\nu - \lambda) - (\mu - \nu)(\theta + 4\lambda) \\ &= -\theta(\lambda - \mu + \mu - \nu + \nu - \lambda) - \frac{2}{\theta}(\nu\lambda^2 + \nu^2\lambda - \mu\lambda^2 - \mu^2\nu) + 4\lambda(\mu - \nu) \\ &= (\mu - \nu)(\theta - 4\lambda), \end{aligned}$$

and the condition is

$$\left\{ -2(\mu - \nu)\left(\theta + \frac{2\mu\nu}{\theta + \lambda}\right), \dots, (\mu - \nu)(\theta - 4\lambda), \dots \mid \frac{\lambda}{\theta + \lambda}, \frac{\mu}{\theta + \mu}, \frac{\nu}{\theta + \nu} \right\}^2 = 0.$$

Writing this in the form

$$\dots - 2\lambda\mu\nu\left(\theta + \frac{2\mu\nu}{\theta + \lambda}\right) \frac{1}{(\theta + \mu)(\theta + \nu)}(\mu - \nu) + \dots = 0,$$

then observing that

$$\frac{\theta + \mu + \nu}{\theta + \lambda} = \frac{2}{\theta}(\lambda + \theta),$$

the first part is

$$\frac{2\lambda\mu\nu(\mu - \nu)(\nu - \lambda)(\lambda - \mu)}{\theta^2(\theta + \lambda)(\theta + \mu)(\theta + \nu)}(\theta + \lambda),$$

and observing that the sum is

$$\sum 2\lambda(\mu - \nu)(\theta^2 + 2\mu\nu) - \lambda\mu\nu \sum \frac{2(\mu - \nu)(\theta + 2\mu\nu)}{\theta + \lambda} = \frac{2}{\theta}\lambda\mu\nu(\mu - \nu)(\nu - \lambda)(\lambda - \mu),$$

the first part is

$$- \frac{2\lambda\mu\nu(\mu - \nu)(\nu - \lambda)(\lambda - \mu)}{\theta^2} \frac{1}{(\theta + \lambda)(\theta + \mu)(\theta + \nu)}.$$

The second part is

$$\frac{2\lambda\mu\nu}{\theta(\theta + \lambda)(\theta + \mu)(\theta + \nu)} \sum (\mu - \nu)(\lambda^2 - \lambda\theta - 2\theta^2),$$

in which the sum is

$$\sum (\mu - \nu)(\lambda^2 - \lambda\theta - 2\theta^2) = 2\lambda^2(\mu - \nu) = -(\mu - \nu)(\nu - \lambda)(\lambda - \mu),$$

so that the second part is

$$- \frac{2\lambda\mu\nu(\mu - \nu)(\nu - \lambda)(\lambda - \mu)}{\theta^2(\theta + \lambda)(\theta + \mu)(\theta + \nu)},$$

and the sum of the two parts is  $= 0$ , which proves the theorem.

**70.** Let  $x_1, y_1, z_1$  be the coordinates of a critic centre, then the equation of the polar in regard to the twofold-centre conic is

$$(-x_1 + y_1 + z_1)x + (x_1 - y_1 + z_1)y + (x_1 + y_1 - z_1)z = 0,$$

and the equation of the conic through the five points is

$$\frac{x_1(y_1 - z_1)}{x} + \frac{y_1(z_1 - x_1)}{y} + \frac{z_1(x_1 - y_1)}{z} = 0,$$

and these equations together determine the remaining two critic centres.

**71.** I remark in passing that the equation of the one-with-twofold centre locus may also be obtained by means of the equations

$$\frac{\lambda}{x} + \frac{\mu}{y} + \frac{\nu}{z} = 0,$$

$$(-x_1 + y_1 + z_1)x + (x_1 - y_1 + z_1)y + (x_1 + y_1 - z_1)z = 0,$$

which determine the remaining two critic centres corresponding to a given critic centre  $(x_1, y_1, z_1)$ ; in fact, in order that the centre  $(x_1, y_1, z_1)$  may be accompanied by a twofold centre the line must touch the conic; and the analytical condition, substituting therein  $(x, y, z)$  in the place of  $(x_1, y_1, z_1)$ , is found to be

$$xyz\{x^3 + y^3 + z^3\} - (yz^2 + y^2z + zx^2 + z^2x + xy^2 + x^2y) + 3xyz = 0;$$

the three lines  $xyz = 0$  are not properly part of the locus, but their appearance may be accounted for without difficulty.

**72.** Assume that the line  $\lambda x + \mu y + \nu z = 0$  passes successively through the points

$$(x = 0, y - z = 0), (y = 0, z - x = 0), (z = 0, x - y = 0),$$

or, what is the same thing, the points  $(0, 1, 1), (1, 0, 1), (1, 1, 0)$ : then (*ante*, No. 56) the critic centres are in all these cases respectively on the conics

$$\frac{\nu}{y} - \frac{\mu}{z} - \frac{2(\nu - \mu)}{x + y + z} = 0,$$

$$\frac{\lambda}{z} - \frac{\nu}{x} - \frac{2(\lambda - \nu)}{x + y + z} = 0,$$

$$\frac{\mu}{x} - \frac{\lambda}{y} - \frac{2(\mu - \lambda)}{x + y + z} = 0,$$

or, as these may be written,

$$-zy + x(y + z) = 0,$$

$$-xy + y(z + x) = 0,$$

$$-yy + z(x + y) = 0,$$

the first of which is a conic touching the lines  $x = 0, y + z = 0$  at the points of intersection with the line  $y - z = 0$ ; and similarly for the other two conics.

**73.** Suppose that the line  $\lambda x + \mu y + \nu z = 0$  passes through the point  $(4, -1, -1)$ , or let  $4\lambda - \mu - \nu = 0$ ; we have  $(\alpha, \beta, \gamma) = (4, -1, -1)$ ; and the critic centres lie on the curve

$$\frac{4}{x} - \frac{1}{y} - \frac{1}{z} - \frac{4}{x + y + z} = 0,$$

that is

$$\frac{4(y+z)}{x(x+y+z)} - \frac{y+z}{yz} = 0,$$

or, as this may be written,

$$(y+z)\{x(x+y+z) - 4yz\} = 0,$$

so that the cubic locus breaks up into the line  $y+z=0$  and into the conic

$$x(x+y+z) - 4yz = 0.$$

**74.** I say that the critic centres lie, one of them on the line, and the other two on the conic. In fact, putting  $\lambda = \frac{1}{4}(\mu + \nu)$  the equation in  $\theta$  is

$$\theta^3 - \theta\mu\nu - 2\lambda\mu\nu = 0,$$

that is

$$(\theta + \frac{1}{2}(\mu + \nu))\{\theta^2 - \frac{1}{2}(\mu + \nu)\theta - \mu\nu\} = 0,$$

and we have

$$x : y : z = \frac{1}{\theta + \lambda} : \frac{1}{\theta + \mu} : \frac{1}{\theta + \nu}.$$

**75.** Hence if  $\theta + \frac{1}{2}(\mu + \nu) = 0$ , we obtain

$$x : y : z = \frac{1}{\frac{1}{4}(\mu + \nu) + \frac{1}{2}(\mu + \nu) + \mu} : \dots,$$

whence also

$$(\mu + \nu)x + (\mu - \nu)y = 0,$$

$$(\mu + \nu)x - (\mu - \nu)z = 0,$$

$$y + z = 0,$$

so that the corresponding critic centre lies on the line  $y+z=0$ ; the last-mentioned equations, restoring the value  $4\lambda$  in place of  $\mu + \nu$ , may also be written

$$4\lambda x + (\mu - \nu)y = 0,$$

$$4\lambda x - (\mu - \nu)z = 0,$$

$$y + z = 0.$$

**76.** If on the other hand

$$\theta^2 - \frac{1}{2}(\mu + \nu)\theta - \mu\nu = 0,$$

or, as this equation may be written,

$$5\theta^2 - (\theta + 2\mu)(\theta + 2\nu) = 0,$$

then observing that in general, in virtue of the equation

$$\frac{1}{\theta + \lambda} + \frac{1}{\theta + \mu} + \frac{1}{\theta + \nu} = \frac{2}{\theta},$$

we have

$$\frac{1}{\theta + \mu} + \frac{1}{\theta + \nu} = \frac{\theta + 2\mu}{\theta(\theta + \mu)},$$



and consequently

$$y : x + z = \theta : \theta + 2\mu, \quad z : x + y = \theta : \theta + 2\nu,$$

the foregoing equation

$$5\theta^2 - (\theta + 2\mu)(\theta + 2\nu) = 0$$

gives

$$5yz - (x + z)(x + y) = 0,$$

that is

$$x(x + y + z) - 4yz = 0;$$

or the critic centres corresponding to the two values of  $\theta$  lie on the conic. The line joining them is the polar of the point  $(4, -1, -1)$  in regard to the twofold-centre conic; the equation therefore is

$$(\mu - \nu)x - (3\mu + \nu)y + (\mu + 3\nu)z = 0.$$

**77.** Starting with a critic centre on the line  $y + z = 0$ , the other two critic centres lie on the conic  $x(x + y + z) - 4yz = 0$ , and they are the intersections of the conic by the polar of the first centre in regard to the twofold-centre conic.

**78.** Starting with a critic centre on the conic  $x(x + y + z) - 4yz = 0$ , the other two critic centres lie one on the conic, and the other on the line  $y + z = 0$ ; viz. the polar of the first centre in regard to the twofold-centre conic meets the line in one point, and the conic in two points; of these one is the harmonic of the point on the line in regard to the twofold-centre conic; this point on the conic, and the point on the line, are the other two centres.

**79.** The point  $(4, -1, -1)$  is of course one of a system of three points; viz. these are  $(4, -1, -1)$ ,  $(-1, 4, -1)$ ,  $(-1, -1, 4)$ ; and the corresponding loci of the critic centres are

$$(y + z)\{x(x + y + z) - 4yz\} = 0,$$

$$(z + x)\{y(x + y + z) - 4zx\} = 0,$$

$$(x + y)\{z(x + y + z) - 4xy\} = 0,$$

the three points in question are (*ante*, No. 24) shown to be nodes of the twofold-centre envelope.

**80.** The line  $5x + y + z = 0$  is the line through the points  $(-1, 4, -1)$ ,  $(-1, -1, 4)$ , and as such the corresponding critic centres lie

- one on the line  $z + x = 0$ , two on the conic  $x(x + y + z) - 4yz = 0$ ,
- one on the line  $x + y = 0$ , two on the conic  $y(x + y + z) - 4zx = 0$ .

The two lines meet in the point  $(1, -1, -1)$ .

The two conics meet in the points  $(1, 0, 0)$ ,  $(2, 3, 3)$ ; and touch at the point  $(0, 1, -1)$ , the common tangent being  $5x + y + z = 0$ : this appears by writing the equations of the two conics in the forms

$$(y - z)(5x + y + z) + (y + z)(-3x + y + z) = 0,$$

$$-(y - z)(5x + y + z) + (y + z)(-3x + y + z) = 0,$$

for we have then the four points of intersection put in evidence; viz. these are

$$y - z = 0, \quad y + z = 0, \quad \text{that is } (1, 0, 0),$$

$$y - z = 0, \quad -3x + y + z = 0, \quad (2, 3, 3),$$

$$5x + y + z = 0, \quad y + z = 0, \quad (0, 1, -1),$$

$$5x + y + z = 0, \quad -3x + y + z = 0, \quad (0, 1, -1).$$

The point of intersection  $(1, 0, 0)$ , which is an angle of the triangle, is not a critic centre; the three critic centres are the other point of intersection  $(2, 3, 3)$ ; the point of contact  $(0, 1, -1)$ ; and the point of intersection  $(1, -1, -1)$  of the two lines.

**81.** To obtain in a different manner the last-mentioned result it may be remarked that for the line  $3x + y + z = 0$ , for which  $(\lambda, \mu, \nu) = (3, 1, 1)$ , the equation in  $\theta$  is

$$\theta^3 - 7\theta - 6 = (\theta + 1)(\theta + 2)(\theta - 3) = 0,$$

so that the values of  $\theta + \lambda$ ,  $\theta + \mu$ ,  $\theta + \nu$  are

$$\begin{aligned} \text{for } \theta &= -1, & 2, & 0, & 0, \\ \theta &= -2, & 1, & -1, & -1, \\ \theta &= 3, & 6, & 4, & 4, \end{aligned}$$

and the corresponding values of  $x : y : z$  are

$$\begin{aligned} \theta &= -1 : \frac{1}{2} : \infty : \infty, \text{ that is } (0, 1, -1), \\ \theta &= -2 : 1 : -1 : -1, \text{ that is } (1, -1, -1), \\ \theta &= 3 : \frac{1}{6} : \frac{1}{4} : \frac{1}{4}, \text{ that is } (2, 3, 3), \end{aligned}$$

which points are therefore the critic centres for the line  $3x + y + z = 0$ .

The last-mentioned line, it is clear, is one of the system of three lines

$$3x + y + z = 0, \quad x + 3y + z = 0, \quad x + y + 3z = 0.$$

**82.** If  $\lambda = 0$ , that is if the line  $\lambda x + \mu y + \nu z = 0$  pass through an angle  $y = 0$ ,  $z = 0$  of the triangle; then reverting to the original equations

$$\frac{-x + y + z}{\lambda x} = \frac{x - y + z}{\mu y} = \frac{x + y - z}{\nu z},$$

these give  $(y = 0, z = 0)$  or else  $(-x + y + z = 0, \nu z^2 - \mu y^2 = 0)$ , that is, one of the three critic centres is the angle  $(y = 0, z = 0)$  of the triangle; and the other two are the intersections of the line  $-x + y + z = 0$  with the pair of lines  $\nu z^2 - \mu y^2 = 0$ .

It should be remarked that, given the critic centre  $y = y_1 = 0$ ,  $z = z_1 = 0$ , the remaining two centres cannot be determined as the intersection of the polar  $-x + y + z = 0$  with the conic

$$\frac{x_1(y_1 - z_1)}{x} + \frac{y_1(z_1 - x_1)}{y} + \frac{z_1(x_1 - y_1)}{z} = 0,$$

inasmuch as the equation of this conic becomes the identity  $0 = 0$ .

**83.** The critic centres for the case in question,  $\lambda = 0$ , may also be determined by means of the equation of the cubic through the three centres; in fact, since  $\lambda = 0$ , the equation  $\lambda\alpha + \mu\beta + \nu\gamma = 0$  becomes  $\mu\beta + \nu\gamma = 0$ , that is  $\beta : \gamma = \nu : -\mu$ ; and the equation of the cubic therefore is

$$\nu \left( \frac{1}{y} - \frac{1}{x + y + z} \right) - \mu \left( \frac{1}{z} - \frac{1}{x + y + z} \right) = 0,$$

and since the ratio  $\alpha : \beta$  is arbitrary we have the two equations

$$\frac{1}{x} - \frac{2}{x + y + z} = 0, \quad \frac{\nu}{y} - \frac{\mu}{z} = 0,$$

which resolve themselves into the above-mentioned two equations,  $-x + y + z = 0$ ,  $\nu z^2 - \mu y^2 = 0$ .

**84.** Consider a critic centre the coordinates of which are  $(0, y_1, z_1)$ , that is, which is an arbitrary point on the side  $x = 0$  of the triangle: it is to be remarked that there is not any position of the line  $\lambda x + \mu y + \nu z = 0$ , which properly gives rise to such a critic centre. For writing  $x_1 = 0$  the equations

$$\frac{-x_1 + y_1 + z_1}{\lambda x_1} = \frac{x_1 - y_1 + z_1}{\mu y_1} = \frac{x_1 + y_1 - z_1}{\nu z_1}$$

give  $\mu = 0, \nu = 0$ , that is, the line  $\lambda x + \mu y + \nu z = 0$  is found to be  $x = 0$ ; but in this case the cubic is  $x(yz + k(x + y + z)^2) = 0$ , which irrespectively of the value of  $k$  has nodes at the points  $x = 0, yz + k(y + z)^2 = 0$ , and which only for the value  $k = 0$  acquires a third node at the point  $y = 0, z = 0$ : the case is a singular and exceptional one.

**85.** If notwithstanding we assume a critic centre at the point  $(0, y_1, z_1)$ , then the other two critic centres are by the general theorem given as the intersection of the line

$$(y_1 + z_1)x - (y_1 - z_1)(y - z) = 0$$

with the conic (pair of lines)  $x(y - z) = 0$ , that is, we have a twofold centre  $x = 0, y - z = 0$ , or what is the same thing a twofold centre  $(0, 1, 1)$ .

**86.** If a critic centre lie on the line  $y - z = 0$ , then of the other two critic centres, one lies on this same line and the other is the point  $x = 0, x + y + z = 0$ , or say the point  $(0, 1, -1)$ . And in this case the line  $\lambda x + \mu y + \nu z = 0$  passes through the last-mentioned point; that is, we have  $\mu = \nu$ . Conversely, starting from the equation  $\mu = \nu$ , so that the line  $\lambda x + \mu y + \nu z = 0$  is  $\lambda x + \mu(y + z) = 0$ , a line through the intersection of the lines  $x = 0, x + y + z = 0$ , the equation in  $\theta$  is

$$(\theta + \mu)(\theta^2 - \mu\theta - 2\lambda\mu) = 0,$$

where the factor  $\theta + \mu = 0$  corresponds to the critic centre  $x = 0, x + y + z = 0$ , or  $(0, 1, -1)$  (it will presently be shown that this is so), and the quadric equation  $\theta^2 - \mu\theta - 2\lambda\mu = 0$  corresponds to two critic centres on the line  $y - z = 0$ . We have

$$x : y : z = \frac{1}{\theta + \lambda} : \frac{1}{\theta + \mu} : \frac{1}{\theta + \mu},$$

and thence  $y - z = 0$ ; and  $\theta(x - y) = -\lambda x + \mu y$ , which substituted in the equation  $\theta^2 - \mu\theta - 2\lambda\mu = 0$  gives

$$(\lambda x - \mu y)\{(\lambda + \mu)x - 2\mu y\} - 2\lambda\mu(x - y)^2 = 0,$$

and the two critic centres are given as the intersections of this conic by the line  $y - z = 0$ .

**87.** Consider for a moment the case  $\nu = \mu + \varepsilon$ , where  $\varepsilon$  is ultimately  $= 0$ . The equation in  $\theta$  is

$$\frac{1}{\theta + \lambda} + \frac{1}{\theta + \mu} + \frac{1}{\theta + \mu + \varepsilon} = \frac{2}{\theta};$$

then if a root is  $\theta = -\mu + A\varepsilon$ , we have

$$\frac{1}{A\varepsilon + \lambda - \mu} + \frac{1}{A\varepsilon} + \frac{1}{(A + 1)\varepsilon} = \frac{2}{A\varepsilon - \mu};$$

so that,  $\varepsilon$  being indefinitely small, we have

$$\frac{1}{A} + \frac{1}{A + 1} = 0, \quad \text{that is } 2A + 1 = 0 \text{ or } A = -\frac{1}{2},$$

and then

$$\theta = -\mu - \frac{1}{2}\varepsilon, \quad \theta + \lambda = \lambda - \mu - \frac{1}{2}\varepsilon, \quad \theta + \mu = -\frac{1}{2}\varepsilon, \quad \theta + \nu = +\frac{1}{2}\varepsilon,$$

which gives

$$x : y : z = \frac{1}{\theta + \lambda} : \frac{1}{\theta + \mu} : \frac{1}{\theta + \nu} = \frac{1}{\lambda - \mu - \frac{1}{2}\varepsilon} : -\frac{2}{\varepsilon} : \frac{2}{\varepsilon},$$

or,  $\varepsilon$  being indefinitely small,  $x : y : z = 0 : 1 : -1$ , so that the factor  $\theta + \mu = 0$  corresponds, as mentioned above, to the critic centre  $(0, 1, -1)$ .

[End of Paper 349. The paper concludes with Article 87 above, completing the lengthy analytical treatment of the involution of cubic curves of the form  $xyz + k(x + y + z)^2(\lambda x + \mu y + \nu z) = 0$ .]

## 350. On the Classification of Cubic Curves

[From the *Transactions of the Cambridge Philosophical Society*, vol. xi. Part i. (1866), pp. 81–128. Read April 18, 1864.]

The notion of a curve of a given order may be considered as arising from Descartes' invention of the method of coordinates; and one of the earliest applications of the method was made by Sir Isaac Newton in the *Enumeratio linearum tertii Ordinis* (1706), a work worthy of its author, and which opened a new field of geometrical science. The classification is according to the nature of the infinite branches; there are fourteen genera containing together seventy-two species, but four species were added by Stirling in his *Lineae tertii Ordinis Newtonianae; sive Illustratio dx.* (1717), and two more by Murdoch or Cramer,<sup>1</sup> making in all seventy-eight species. A new classification was made by Plücker in his *System der Analytischen Geometrie*, 1835; this is likewise according to the nature of the infinite branches, but after his six head divisions, and some subordinate divisions thereof, Plücker establishes the divisions called *Groups*, which have nothing analogous to them in the Newtonian theory; there are sixty-one groups, and the total number of species is 219.

The present Memoir contains an exposition of the foregoing classifications, and of the principles on which they are founded, in so far as relates to the superior divisions of the two classifications; and in particular I develop more completely than was done by Plücker the theory of the division into groups. I do not however consider otherwise than very slightly the ultimate division into species.

The above-mentioned work of Newton contains, under the heading “Genesis Curvarum per Umbras,” the remarkable theorem that the curves of the third order may all of them be considered as the shadows of the five Divergent Parabolas; I reserve for a separate Memoir the whole series of considerations to which this theorem gives rise.

I commence by establishing the theory of the classification of cubic curves according to the nature of their infinite branches, in what appears to me the scientifically correct manner as follows.

*The Seven Head Divisions, Article Nos. 1 to 4.*

1. A line in general, and therefore the line Infinity, meets a cubic curve in three points, and these may be

- Three onefold points,
- A twofold point and a onefold (or, as it may also be termed, a one-with-twofold) point,
- A threefold point.

2. But in the second case the line Infinity

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<sup>1</sup>The two additional species are, I believe, first mentioned in Murdoch's *Genesis Curvarum per Umbras* (1746), but one of them is there ascribed to Cramer.

- may be a proper tangent to the curve,
- may pass through a node,
- may pass through a cusp;

and in the third case the line Infinity

- may touch the curve at an inflexion,
- may at a node touch one of the two branches,
- may touch the curve at a cusp.

3. The first case, the three divisions of the second case, and the three divisions of the third case, give in all seven divisions, which, as will appear in the sequel, fall in with Newton's classification, and can be named in his language, viz.

|                                     |                             |
|-------------------------------------|-----------------------------|
| Three onefold points,               | The Hyperbolas.             |
| A onefold and a twofold point;      |                             |
| Infinity a proper tangent.          | The Parabolic Hyperbolas.   |
| Do. through a node.                 | The Central Hyperbolisms.   |
| Do. through a cusp.                 | The Parabolic Hyperbolisms. |
| A threefold point;                  |                             |
| Infinity a tangent at an inflexion. | The Divergent Parabolas.    |
| Do. Do. at a node, to one branch.   | The Trident Curve.          |
| Do. Do. at a cusp.                  | The Cubical Parabola.       |

4. As regards the signification of these terms, it may be remarked that the Hyperbolas have hyperbolic branches, the Parabolic Hyperbolas, hyperbolic and parabolic branches; where by a hyperbolic branch is meant one having an asymptote, and by a parabolic branch one not having an asymptote. The hyperbolism of any curve is the curve derived from it by altering the ordinate in the ratio of the abscissa to any given line

$$(y : y' = x : a, \text{ or say } y' = \frac{ay}{x});$$

the expression Central Hyperbolism is used to include Newton's hyperbolisms of the hyperbola and ellipse; and the expression Parabolic Hyperbolism to denote his hyperbolism of the parabola. The Divergent Parabolas are curves the branches of which ultimately diverge from each other as in the semicubical parabola  $y^2 = x^3$ . The names Trident Curve and Cubical Parabola are in fact not generic but specific; it so happens that the genera to which they respectively belong contain each only a single species. The names for the several kinds of curves are not scientifically devised ones, but it is convenient to have them such as they are.

The foregoing seven divisions, uniting in one the Central Hyperbolisms and the Parabolic Hyperbolisms, are the six head divisions of Plücker.

*Asymptotes, &c. Equations for the Seven Head Divisions. Article Nos. 5 to 22.*

5. For a Hyperbola there is at each of the points at infinity a tangent, which is an asymptote; and the hyperbola has thus three asymptotes.

6. For a Parabolic Hyperbola there is at the onefold point at infinity a tangent, which is an asymptote. There may be described a conic having with the curve at the twofold point at infinity a five-pointic intersection.<sup>2</sup> Such conic, as having the line infinity for a tangent, is a parabola, and it may be termed the asymptotic parabola: the Parabolic Hyperbola has thus an asymptote and an asymptotic parabola.

<sup>2</sup>I have elsewhere spoken of the conic of five-pointic contact: the expression five-pointic intersection is more accurate.

7. For a Central Hyperbolism there is at the onefold point at infinity a tangent which is an asymptote, and which for distinction may be called the onefold asymptote; and at the node or twofold point at infinity there is a pair of tangents which are the parallel asymptotes.

8. For a Parabolic Hyperbolism there is at the onefold point at infinity a tangent which is an asymptote, and which may be called the onefold asymptote; and at the cusp or twofold point at infinity a twofold tangent which is an asymptote, and which may be called the twofold asymptote.

9. For a Divergent Parabola there is not any asymptote or asymptotic conic; but we may consider an asymptotic cubic, viz. this will be a semicubical parabola ( $y^2 = x^3$ ), which is in fact one of the divergent parabolas, the cuspidal divergent parabola, and which may be in general so determined as to have at the inflexion or threefold point at infinity a seven-pointic intersection. For the asymptotic cubic, in order that it may have a cusp, must satisfy two conditions; it may therefore be made to satisfy seven more conditions, or to have a seven-pointic intersection; and then the original curve having an inflexion or threefold point at infinity, the asymptotic cubic will *ipso facto* have the same point as an inflexion, or threefold point at infinity, and be thus a cuspidal divergent parabola.

10. For the Trident Curve, we have at the node or threefold point at infinity, viz. to the branch which is not touched by the line infinity, a tangent which is an asymptote: this cuts at the node the other branch of the curve, and it is therefore an asymptote of three-pointic intersection. We may describe a conic having at the node a five-pointic intersection with the other branch of the curve; such conic as touching the line infinity is a parabola, and it may be called the asymptotic parabola; since the parabola cuts at the node the first-mentioned branch of the curve, viz. the branch not touched by the line infinity, the parabola is in fact a parabola of six-pointic intersection. The Trident Curve has thus an asymptote and an asymptotic parabola of six-pointic intersection.

11. For the Cubical Parabola there is not any asymptote or asymptotic conic: the curve *quâ* curve having a cusp (viz. the cusp or threefold point at infinity) has a single inflexion; and the line joining the cusp with the inflexion, regarded as a threefold line, has with the curve a six-pointic intersection at infinity, and may be considered as an asymptotic cubic.

12. We have in every case a cubic curve  $V = 0$  having with the original curve an intersection at infinity which is at least six-pointic, and which I call the asymptotic aggregate: viz. the asymptotic aggregate is

- For the Hyperbolas; the three asymptotes, intersection six-pointic.
- For the Parabolic Hyperbolas; the asymptote and the asymptotic parabola, intersection seven-pointic.
- For the Central Hyperbolisms; the onefold asymptote and the parallel asymptotes, intersection eight-pointic.
- For the Parabolic Hyperbolisms; the onefold asymptote and the twofold asymptote regarded as a twofold line; intersection eight-pointic.
- For the Divergent Parabolas; the asymptotic semicubical parabola, intersection seven-pointic.
- For the Trident Curve; the asymptote and the asymptotic parabola, intersection nine-pointic.
- For the Cubical Parabola; the line joining the cusp at infinity with the inflexion, regarded as a threefold line, intersection six-pointic.

**13.** I have said that the intersection at infinity is at least six-pointic; but more than this, the intersection at any onefold point at infinity is at least two-pointic; at a twofold point at infinity it is at least four-pointic; and at a threefold point at infinity it is at least six-pointic.

It follows that the intersections at infinity of the cubic and the asymptotic aggregate include the six intersections of the cubic by the line infinity considered as a twofold line; and hence the remaining three intersections of the cubic and the asymptotic aggregate must lie in a line  $s = 0$  (Plücker's line  $S$ ), which I call the *satellite line*. And writing  $z = 0$  for the equation of the line infinity, the equation of the cubic is of the form  $U = V + \mu z^2 s = 0$ . It is to be observed moreover, that when, as for the Hyperbolas and the Cubical Parabola, the intersection at infinity is six-pointic, the line  $s = 0$  is an arbitrary line; when as for the Parabolic Hyperbolas, and the Divergent Parabolas, the intersection is seven-pointic, the line  $s = 0$  meets the cubic in a given point at infinity, viz. the twofold or the threefold point at infinity; and when as for the Central Hyperbolisms and the Parabolic Hyperbolisms the intersection is eight-pointic, the line  $s = 0$  has with the cubic a given twofold intersection at infinity; this however merely implies that the line  $s = 0$  passes through the node or cusp at infinity, and so imposes only one condition on the line  $s = 0$ . Finally, when as in the Trident Curve the intersection is nine-pointic, the line  $s = 0$  has with the curve a given threefold intersection at infinity; that is, it coincides with the line infinity,  $z = 0$ .

**14.** The preceding considerations in regard to the asymptotic aggregate  $V = 0$ , lead very directly to the best analytical form of the function  $V$ , and therefore to that of the equation  $U = V + \mu z^2 s = 0$ , of the cubic.

**15.** For the Hyperbolas; the equations of the asymptotes being  $p = 0$ ,  $q = 0$ ,  $r = 0$ , we have  $V = pqr = 0$  for the asymptotic aggregate; the satellite line is arbitrary, and hence

$$\text{Equation of the Hyperbolas is } pqr + \mu z^2 s = 0.$$

**16.** For the Parabolic Hyperbolas. Imagine parallel to the asymptote a line  $q = 0$  touching the asymptotic parabola; and let the line joining the point of contact with the twofold point at infinity have for its equation  $q = 0$ ; the equation of the asymptote is

$$p + \kappa z = 0,$$

that of the asymptotic parabola is  $q^2 + \lambda pz = 0$ , and hence the equation of the asymptotic aggregate is  $(p + \kappa z)(q^2 + \lambda pz) = 0$ ; the satellite line passes through the twofold point at infinity, or its equation is  $q + \sigma z = 0$ ; hence

$$\text{Equation of the Parabolic Hyperbolas is } (p + \kappa z)(q^2 + \lambda pz) + \mu z^2(q + \sigma z) = 0.$$

**17.** For the Central Hyperbolisms; the equation of the onefold asymptote is taken to be  $p = 0$ , and that of the parallel asymptotes to be  $q^2 + \kappa z^2 = 0$ ; hence the equation of the asymptotic aggregate is  $p(q^2 + \kappa z^2) = 0$ , the satellite line passes through the twofold point at infinity, its equation is  $q + \sigma z = 0$ ; hence

$$\text{Equation of the Central Hyperbolisms is } p(q^2 + \kappa z^2) + \mu z^2(q + \sigma z) = 0.$$

**18.** For the Parabolic Hyperbolisms: the only difference is that instead of the parallel asymptotes  $q^2 + \kappa z^2 = 0$  we have the twofold asymptote  $q^2 = 0$ ; hence

$$\text{Equation of the Parabolic Hyperbolisms is } pq^2 + \mu z^2(q + \sigma z) = 0.$$

**19.** For the Divergent Parabolas: the asymptotic aggregate is a semicubical parabola; let  $q = 0$  be the equation of the cuspidal tangent,  $p = 0$  the equation of the line joining the cusp

with the inflexion at infinity, then the equation is  $p^3 + \lambda q^2 z = 0$ . The satellite line passes through the threefold point at infinity, its equation is  $p + \sigma z = 0$ ; hence

$$\text{Equation of the Divergent Parabolas is } p^3 + \lambda q^2 z + \mu z^2(p + \sigma z) = 0.$$

**20.** For the Trident Curve: let  $p = 0$  be the equation of the asymptote,  $q = 0$  that of the tangent to the asymptotic parabola at the point not at infinity where it is met by the asymptote, then the equation of the parabola is  $p^2 + \lambda qz = 0$ , and that of the asymptotic aggregate is  $p(p^2 + \lambda qz) = 0$ ; the satellite line is the line infinity,  $z = 0$ ; hence

$$\text{Equation of the Trident Curve is } p(p^2 + \lambda qz) + \mu z^3 = 0.$$

**21.** For the Cubical Parabola: let  $p = 0$  be the equation of the line joining the inflexion with the cusp at infinity, then the asymptotic aggregate is this line taken as a threefold line, or the equation is  $p^3 = 0$ ; the satellite line is arbitrary; hence

$$\text{Equation of the Cubical Parabola is } p^3 + \mu z^2 s = 0.$$

**22.** It is convenient to notice here that for the Hyperbolas the line  $s = 0$  is determined as follows, viz. the line infinity meets the curve in three points, and the tangents at these points (the asymptotes) again meet the curve in three points lying in a line which is the line in question; in other words, the line  $s = 0$  is (in the sense in which I have elsewhere used the term) the satellite line of infinity. For the other kinds of cubic curves, the line  $s = 0$  is not, in the sense just referred to, the satellite line of infinity; but in the present Memoir I shall in every case call the line  $s = 0$ , the satellite line.

*The Thirteen Divisions. Article Nos. 23 to 33.*

**23.** The characters of the foregoing seven divisions are irrespective of reality; and before going further it may be remarked, that as to the Hyperbolas and the Parabolic Hyperbolas a subdivision also irrespective of reality may be made as follows.

**24.** For a Hyperbola, the three asymptotes may not meet in a point, or they may meet in a point. For shortness I say that in the former case we have a Hyperbola *A*, in the latter case a Hyperbola *B*. I consider more particularly (post. No. 41) the special case of a Hyperbola *B*.

**25.** For a Parabolic Hyperbola, the asymptote may meet the asymptotic parabola in two onefold points; or in a twofold point.

**26.** I come now to the divisions which depend on reality: it is assumed that the curve is real.

**27.** For the Hyperbola the three points at infinity may be all real or else one real, two imaginary. In the former case, the asymptotes are all real, and we have the *redundant hyperbola*; in the latter case the real point at infinity gives rise to a real asymptote, the imaginary points to imaginary asymptotes: we have in this case the *defective hyperbola*. It is to be noticed that the imaginary asymptotes meet in a real point, called the *asymptote-point*; and that such point, if we regard it as an indefinitely small ellipse given as to the position and ratio of its axes, determines the imaginary asymptotes. Combining the division with the *A*, *B*, we have four subdivisions of the Hyperbola.

**28.** For a Hyperbola *A* redundant the three asymptotes form a triangle, and for a Hyperbola *B* redundant they meet in a point. For a Hyperbola *A* defective, the asymptote-point does not lie on the real asymptote; for a Hyperbola *B* defective it does lie on the real asymptote.

**29.** For a Parabolic Hyperbola: the onefold point and the twofold point at infinity are of necessity real, as are also the asymptote and the asymptotic parabola. If the asymptote meets



the asymptotic parabola in two onefold points, these may be both real or both imaginary; if it meets it in a twofold point, this is real. We have thus three subdivisions of the Parabolic Hyperbola. For the Central Hyperbolism, the onefold point, and the node or twofold point at infinity, are both real; the asymptote is also real. But the node may be a crunode or an acnode; that is, the tangents at the node, or parallel asymptotes, may be both real, or both imaginary: we have thus two subdivisions, viz. the Hyperbolism of the hyperbola, and the Hyperbolism of the ellipse.

**30.** For the Parabolic Hyperbolism, the onefold point and the cusp or twofold point at infinity, and also the onefold asymptote and the twofold asymptote are all real.

**31.** For the Divergent Parabola, the inflexion or threefold point at infinity is real.

**32.** For the Trident Curve the node or threefold point at infinity is real, and inasmuch as one of the tangents is the line infinity, the node is a crunode, and the other tangent, or asymptote of the curve, is also real.

**33.** For the Cubical Parabola, the cusp or threefold point at infinity and the tangent at this point are each real.

Reckoning the hyperbolas as 4, the parabolic hyperbolas as 3, the central hyperbolisms as 2, and the parabolic hyperbolisms, the divergent parabolas, the trident curve, and the cubical parabola, each as 1, we have in all 13 divisions.

*The Notion of a Group. Article No. 34.*

**34.** I remark that the characters as well of the 7 divisions as of the 13 divisions have exclusive reference to the form of the asymptotic aggregate  $V = 0$ ; we have an ulterior division depending on the relation of the satellite line to the asymptotic aggregate, and which I regard as the proper origin of Plücker's Groups: viz. for a given form of the asymptotic aggregate  $V = 0$ , and corresponding to each characteristically distinct position in relation thereto of the satellite line  $s = 0$ , we have a *Group*. The determination of the characteristically distinct positions of the satellite line cannot be completely effected *à priori*; for instance, in the case of the Hyperbolas *A* redundant, the distinctions which immediately present themselves are that the satellite line cuts the three sides produced, or two sides and the third side produced, of the triangle formed by the asymptotes, or passes through an angle of the triangle, &c.; but these are not all the distinctions which have to be made; to determine them, taking the satellite line as given, we discuss the series of curves represented by the equation  $V + \mu z^2 s = 0$ ; for instance (and it is on this that the discussion chiefly turns), we see that the parameter  $\mu$  may be so determined that the curve shall have a node, but the reality or non-reality of the roots of the equation in  $\mu$ , and therefore the existence of a real nodal curve or curves, will depend on the position of the satellite line  $s = 0$ ; and it is thus only by the discussion of the group that we arrive at an enumeration of the different groups.

*Osculating Asymptotes and other Specialities. Article Nos. 35 to 41.*

**35.** But Plücker nevertheless, prior to the establishment of his groups, introduces certain intermediate divisions as to osculating asymptotes, &c., which have really reference to the position of the satellite line; an osculating asymptote gives rise to a 'diameter,' and the diameter is a distinctive character in the Newtonian genera; to explain how all this is, I proceed as follows.

**36.** The parallel asymptotes of a Central Hyperbolism, the twofold asymptote of a Parabolic Hyperbolism and the asymptote of the Trident Curve are singular asymptotes, that is, each of them touches the curve at a node or a cusp, and is thus an asymptote of three-pointic intersection. Excluding these, and using the term asymptote to denote a non-singular asymptote, an asymptote is in general an ordinary tangent or asymptote of two-pointic intersection; if, however, the point of contact is an inflexion, then the asymptote is an asymptote of three-pointic intersection, or osculating asymptote. In particular for the Hyperbolas, the asymptotes

may be all ordinary, or they may be two ordinary and one osculating, or all three osculating; but they cannot be only two of them osculating; for the line through two inflexions meets a cubic curve in a third point which is also an inflexion; that is, if two asymptotes are osculating, the third is also an osculating asymptote. The foregoing remarks apply as well to the defective as the redundant Hyperbolas; it is to be noticed, however, as regards the defective Hyperbolas that the osculating asymptote, when there is only one, is necessarily the real asymptote, and consequently that the cases are – asymptotes ordinary; the real asymptote alone osculating; three osculating asymptotes. For the Parabolic Hyperbolas the asymptote, and for the Central Hyperbolisms and the Parabolic Hyperbolisms the onefold asymptote, may be ordinary or osculating.

**37.** The distinction of ordinary and osculating asymptotes has reference to the position of the satellite line; viz. for the Hyperbolas, when there is a single osculating asymptote, the satellite line passes through the point at infinity of the osculating asymptote, or what is the same thing, the satellite line is parallel to the osculating asymptote; and when there are three osculating asymptotes, the satellite line coincides with the line infinity. And, conversely, when the satellite line is parallel to an asymptote such asymptote is an osculating one, and when the satellite line is at infinity the three asymptotes are osculating. For the Parabolic Hyperbolas the asymptote, and for the Hyperbolisms the onefold asymptote, is an osculating asymptote when the satellite line is at infinity; and conversely.

**38.** There is in regard to the Divergent Parabolas a distinction which may be mentioned here; viz. the satellite line may disappear altogether ( $\mu = 0$ ), and the curve thus coincide with the asymptotic semicubical parabola. Or, what is the general case, the satellite line may be distinct from the line infinity, – and it may cut in two real points, touch, or cut in two imaginary points the asymptotic semicubical parabola; or the satellite line may coincide with the line infinity, the asymptotic semicubical parabola being in this case of nine-pointic intersection.

**39.** The term “diameter” is used by Newton in the *Enumeratio* in two different senses; viz. for any given direction of the ordinates there exists a right line or “diameter,” such that measuring the ordinates from this line the sum  $y + y' + y''$  of the three ordinates is  $= 0$ . Such diameter is in fact the second or line polar in regard to the cubic of an arbitrary point on the line infinity. But the term diameter is afterwards and will be here used to denote a diameter *absolute dictum*, viz. for a direction of the ordinates parallel to a non-singular asymptote there may exist a right line or “diameter” such that the ordinates measured from this point are equal and opposite to each other, or what is the same thing, such that the sum  $y + y'$  of the two ordinates is  $= 0$ ; this implies that the asymptote is an osculating asymptote. In fact, the first or conic polar of any inflexion of the cubic breaks up into a pair of lines, one of which is the tangent at the inflexion, the other of them, the ‘polar’ of the inflexion, a line which cuts harmonically the chords through the inflexion, and which when the inflexion is at infinity becomes a diameter. The remarks previously made as to osculating asymptotes apply therefore to diameters, viz. the Hyperbolas may have no diameter, a single diameter, or three diameters, &c.

**40.** Newton speaks also of the “centre” of a cubic curve; viz. there may be a point on the curve such that for any line through this point the two radius vectors are equal and opposite to each other, or that the sum  $r + r'$  of the two radius vectors is  $= 0$ . The centre is in fact a point of inflexion which has for its polar the line infinity. The curves which may have a centre are the Hyperbolas (redundant or defective), the Central Hyperbolisms (of the hyperbola or ellipse) and the Cubical Parabola. For the hyperbolas, the three asymptotes and the satellite line must meet in a point of the curve, which point is then the centre; for the central hyperbolisms the onefold asymptote and the satellite line must meet in a point of the curve, which point is then a centre; and for the cubical parabola no condition is required, but the inflexion is a centre. I remark here, in passing, that the notion of a centre as just explained has no place in Plücker’s Classification, and that the two Newtonian species 58 and 59 (hyperbolisms of the hyperbola) and the two Newtonian species 61 and 62 (hyperbolisms of the ellipse) which differ, the two of

a pair from each other, according as there is no centre or a single centre, form each pair a single species with Plücker; viz. they are 198 and 201 respectively.

**41.** It has been already remarked that the three asymptotes of a Hyperbola may meet in a point. As to this it is to be noticed that from any point we may draw six tangents to a cubic, the points of contact lie on a conic, the conic polar of the point; if, however the point lie on the Hessian of the cubic, then the conic breaks up into a pair of lines, each of which is a tangent to the Pippian; the two lines meet in a point of the Hessian, which point forms with the first mentioned point a pair of conjugate poles of the cubic.<sup>3</sup>

Conversely, any tangent of the Pippian meets the cubic in three points, the tangents at which meet in a point of the Hessian; and from this point we may draw to the cubic three other tangents the points of contact of which lie on a line which is also a tangent of the Pippian, and the two tangents of the Pippian meet in a point of the Hessian; the two points of the Hessian being conjugate poles of the cubic. In particular, if the line infinity is a tangent of the Pippian, then the three asymptotes meet in a point of the Hessian, and the three tangents from this point to the cubic touch the cubic in three points lying on a line which is a tangent of the Pippian, and which meets the line infinity in a point forming with the first mentioned point a pair of conjugate poles of the cubic.

I proceed now to explain the classification of Newton so far as relates to the division into genera, and the classification of Plücker so far as relates to the divisions immediately superior to the groups.

*Newton's Classification. Article Nos. 42 to 46.*

**42.** Newton establishes in the first instance the following four cases; viz. the equation of a cubic curve is one of the forms

$$\begin{aligned} \text{I.} \quad & xy^2 + ey = ax^3 + bx^2 + cx + d, \\ \text{II.} \quad & xy = ax^3 + bx^2 + cx + d, \\ \text{III.} \quad & y^2 = ax^3 + bx^2 + cx + d, \\ \text{IV.} \quad & y = ax^3 + bx^2 + cx + d. \end{aligned}$$

It is not, I think, necessary to reproduce here the very interesting reasoning by means of which this most important step in the classification was effected.

**43.** Starting from the four cases, Newton obtains his 14 genera, viz. Case I gives 11 genera, and Cases II, III, IV give each a single genus. But these genera group themselves as follows, viz. 1, 2, 3, 4, 5, 6 are Hyperbolas; 7 and 8, Parabolic Hyperbolas; 9 and 10, Central Hyperbolisms; 11, Parabolic Hyperbolisms; 12, the Trident Curve; 13, the Divergent Parabolas; and 14, the Cubical Parabola. And the equations are as follows:

$$\begin{aligned} \text{the Hyperbolas} \quad & xy^2 + ey = ax^3 + bx^2 + cx + d, \\ \text{the Parabolic Hyperbolas} \quad & xy^2 + ey = bx^2 + cx + d, \\ \text{the Central Hyperbolisms} \quad & xy^2 + ey = cx + d, \\ \text{the Parabolic Hyperbolisms} \quad & xy^2 + ey = d, \\ \text{the Divergent Parabolas} \quad & y^2 = ax^3 + bx^2 + cx + d, \\ \text{the Trident Curve} \quad & xy = ax^3 + bx^2 + cx + d, \\ \text{the Cubical Parabola} \quad & y = ax^3 + bx^2 + cx + d, \end{aligned}$$

where it is to be understood that the highest expressed power on the right-hand side of each equation does not vanish.

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<sup>3</sup>See as to this theory my *Memoir on Curves of the Third Order*, Phil. Trans. p. 147 (1856), [145].

44. In these equations the axes  $x = 0$ ,  $y = 0$  are not for the most part lines precisely determined in relation to the curve, but it is easy to see as well analytically as geometrically how by a proper transformation of the equations they may be brought into forms such as those previously obtained, in which the several lines  $p = 0$ , etc., stand in a determinate relation to the curve. Thus, taking the equation of the Cubical Parabola, this may be written  $y = a(x + \frac{b}{3a})^3 + c'x + d'$ ; or, what is the same thing,  $y' = ax'^3$ . Or geometrically, we see that  $x = 0$  is a line completely determined as to its direction, it is in fact a line through the cusp at infinity; but that  $y = 0$  is an arbitrary line in regard to the curve; taking for  $y = 0$  the tangent at the inflexion, and for  $x = 0$  the line from the inflexion to the cusp at infinity, then the curve must pass through the point  $(x = 0, y = 0)$ , and  $y = 0$  must give a threefold value of  $x$ ; the equation thus is  $y = ax^3$ . And so in other cases.

45. In the division into genera, Newton distinguishes the Hyperbolas into the redundant and defective, and the redundant hyperbolas into those for which the asymptotes form a triangle, and those for which the asymptotes meet in a point. The redundant hyperbolas with asymptotes forming a triangle are distinguished according as they have no diameter, a single diameter, or three diameters. The like distinction might have been, but is not, made as to the redundant hyperbolas with asymptotes meeting in a point; these are in fact included in a single genus; but the distinction presents itself in the species of that genus. As to the defective hyperbolas, Newton attends only to the real asymptote; and the only distinction is according as they have no diameter or a real diameter. The Parabolic Hyperbolas are in like manner divided according as they have no diameter or a diameter. The Central Hyperbolisms, according as the parallel asymptotes are real or imaginary, are the hyperbolisms of the hyperbola or of the ellipse. The hyperbolisms of the hyperbola form a single genus. Each of the Hyperbolisms might have been distinguished according as there is no diameter or a single diameter; this distinction appears in the species. The Trident Curve, the Divergent Parabolas, and the Cubical Parabola, form each a single genus.

46. We have thus the following Table of the Newtonian genera: I show in it the species in each genus, retaining Newton's numbers, and distinguishing by the numbers 10', 13', 22', 22'' the four species added by Stirling, and by 56' and 56'' the two species added by Murdoch or Cramer: I show also the division of genus 4, according to the number of diameters; and I also show the five species of curves having a centre.

#### Table of the Newtonian Genera.

1. Redundant Hyperbolas with asymptotes forming a triangle, and without a diameter.  
Sp. 1, 2, 3, 4, 5, 6, 7, 8, 9.
2. Redundant Hyperbolas with asymptotes forming a triangle, and with a single diameter.  
Sp. 10, 10', 11, 12, 13, 13', 14, 15, 16, 17, 18, 19, 20, 21.
3. Redundant Hyperbolas with asymptotes forming a triangle, and with three diameters.  
Sp. 22, 22', 22'', 23.
4. Redundant Hyperbola with asymptotes meeting in a point.  
Without a diameter, Sp. 24, 25, 26, 27. With one diameter, Sp. 28, 29, 30, 31. With three diameters, Sp. 32. Sp. 27 has a centre.
5. Defective Hyperbolas without a diameter.  
Sp. 33, 34, 35, 36, 37, 38. Sp. 38 has a centre.
6. Defective Hyperbolas with a diameter.  
Sp. 39, 40, 41, 42, 43, 44, 45.
7. Parabolic Hyperbolas without a diameter.  
Sp. 46, 47, 48, 49, 50, 51, 52.
8. Parabolic Hyperbolas with a diameter.  
Sp. 53, 54, 55, 56, 56', 56''.

**9. Hyperbolisms of the hyperbola.**

Without a diameter, Sp. 57, 58, 59. With a diameter, Sp. 60. Sp. 59 has a centre.

**10. Hyperbolisms of the ellipse.**

Without a diameter, Sp. 61, 62. With a diameter, Sp. 63. Sp. 62 has a centre.

**11. Hyperbolisms of the parabola.**

Without a diameter, Sp. 64. With a diameter, Sp. 65.

**12. Trident Curve, Sp. 66.****13. Divergent Parabolas. Sp. 67, 68, 69, 70, 71.****14. Cubical Parabola. Sp. 72.**

*Plücker's Classification. Article Nos. 47 to 49.*

**47.** Plücker in the first instance obtains by analytical considerations his six head divisions, corresponding to the seven divisions in the present Memoir, the Central Hyperbolisms and the Parabolic Hyperbolisms forming with him a single division. The equations are obtained in the form already mentioned, the only difference being that he writes  $z = 1$ ; his six head divisions with their equations thus are

$$\begin{aligned} \text{Hyperbolas} \quad pqr + \mu s &= 0, \\ \text{Parabolic Hyperbolas} \quad (p + \kappa)(q^2 + \lambda p) + \mu(q + \sigma) &= 0, \\ \text{Hyperbolisms} \quad p(q^2 + \kappa) + \mu(q + \sigma) &= 0, \\ \text{Divergent Parabolas} \quad p^3 + \lambda q^2 + \mu(p + \sigma) &= 0, \\ \text{Trident Curve} \quad p(p^2 + \lambda q) + \mu &= 0, \\ \text{Cubical Parabola} \quad p^3 + \mu s &= 0. \end{aligned}$$

**48.** He then divides the Hyperbolas into the redundant and defective. The redundant hyperbolas are then divided as they have no osculating asymptote, one osculating asymptote, or three osculating asymptotes; and each of these according as the asymptotes form a triangle or meet in a point. As regards the defective hyperbolas he attends to the imaginary asymptotes, represented by means of their real point of intersection, the “asymptote-point,” and the division is thus similar to that of the redundant hyperbolas, viz. the defective hyperbolas are distinguished according as they have no osculating asymptote, a real osculating asymptote, or three osculating asymptotes; and each of these according as the asymptotes form a triangle or meet in a point; that is, according as the asymptote-point does not, or does, lie on the real asymptote.

The Parabolic Hyperbolas are distinguished according as the asymptote is ordinary and the asymptotic parabola one of five-pointic intersection; or, as the asymptote is osculating and the parabola one of six-pointic intersection; and each of these according as the asymptote cuts, touches, or does not cut, the parabola. The Hyperbolisms are distinguished into those of the hyperbola, ellipse, and parabola, and each of these according as the asymptote is ordinary or osculating. The Divergent Parabolas are distinguished in the manner already mentioned; viz. according as the curve is the semicubical parabola; or, as there is a satellite line not at infinity, and an asymptotic semicubical parabola of seven-pointic intersection, and which is cut, touched, or not cut, by the satellite line; or, as the satellite line is at infinity, and the semicubical parabola is of nine-pointic intersection. The Trident Curve and the Cubical Parabola are not divided.

**49.** I annex the following Table showing the Groups included in each division: for shortness I use the before-mentioned symbols  $A, B$  to denote that the asymptotes form a triangle or meet in a point respectively.

**Table of the Plückerian Divisions.**

*Hyperbolas:*

Redundant;  
 No osculating asymptote,  
 A I, II, III, IV, V, VI  
 B VII, VIII  
 One osculating asymptote,  
 A IX, X, XI, XII, XIII, XIV  
 B XV  
 Three osculating asymptotes,  
 A XVI  
 B XVII  
 Defective;  
 No osculating asymptote,  
 A XVIII, XIX, XX, XXI, XXII, XXIII  
 B XXIV, XXV, XXVI, XXVII  
 Real osculating asymptote,  
 A XXVIII, XXIX, XXX, XXXI, XXXII, XXXIII  
 B XXXIV  
 Three osculating asymptotes,  
 A XXXV  
 B XXXVI

*Parabolic Hyperbolas:*

Ordinary asymptote and five-pointic parabola.  
 asymptote cuts parabola XXXVII, XXXVIII, XXXIX, XL, XLI  
 does not cut „ XLII  
 touches „ XLIII, XLIV

Osculating asymptote and six-pointic parabola.  
 asymptote cuts parabola XLV  
 does not cut „ XLVI  
 touches „ XLVII

*Hyperbolisms:*

Of hyperbola;  
 asymptote ordinary XLVIII, XLIX  
 „ osculating L  
 Of ellipse;  
 asymptote ordinary LI  
 „ osculating LII  
 Of parabola;  
 asymptote ordinary LIII  
 „ osculating LIV

*Divergent Parabolas:*

Semicubical Parabola LV  
 Asymptotic ditto, seven-pointic.  
 satellite line cuts semic. parab. LVI  
 does not cut „ LVII  
 touches (i.e. passes through the vertex of) LVIII  
 Asymptotic ditto, nine-pointic LIX

*Trident Curve* LX

*Cubical Parabola* LXI

*As to the Theory of Groups. Article No. 50.*

**50.** It has been already remarked that the division into groups depends on the different positions of the satellite line in regard to the asymptotic aggregate, and that the investigation relates in a great measure to the discussion of the values of the parameter  $\mu$  in the equation  $V + \mu z^2 s = 0$ , which are such as to give rise to a nodal curve. It is to be noticed that except for the Hyperbolas and for the Cubical Parabola, the satellite  $s = 0$  is either a line passing through a given point at infinity (determinate, that is, as regards its direction), or in the case of the Trident Curve it is the given line infinity; there is at most only a single series of positions to be considered, and the theory is a short and easy one; and for the Cubical Parabola, although the satellite line  $s = 0$  is here an arbitrary line, yet on account of the cusp at infinity, there is not any critic value of  $\mu$ , or in fact any distinction of cases. The only other case where the satellite line  $s = 0$  is an arbitrary line, admitting therefore of a double series of positions, is that of the Hyperbolas; and the division into groups constitutes an extensive and interesting theory, which is insufficiently discussed by Plücker; and it was with a view to the development of this theory that my Memoir, *On a Case of the Involution of two Cubic Curves*, (*ante*, pp. 39 to 81), referred to in the sequel as Memoir on Involution, [349], was written. I remark that of the three curves there established as material to the theory, and which are further spoken of in the sequel of the present memoir, viz. the envelope, the twofold-centre locus, and the one-with-twofold-centre locus, Plücker considers only the twofold-centre locus. I proceed to apply the results of that Memoir to the present theory.

*As to the Groups of the Hyperbolas A. Article Nos. 51 to 53.*

**51.** The assumed form of equation was  $pqr + \mu z^2 s = 0$ , but using now

$$x = 0, \quad y = 0, \quad z = 0$$

(instead of  $p = 0, q = 0, r = 0$ ) for the equations of the asymptotes, we may imagine the implicit constants so determined that the line infinity (before represented by  $z = 0$ ) shall have for its equation  $x + y + z = 0$ ; writing moreover  $\lambda x + \mu y + \nu z = 0$  for the satellite line  $s = 0$ , and  $k$  in the place of  $\mu$ , the equation becomes

$$xyz + k(x + y + z)^2(\lambda x + \mu y + \nu z) = 0,$$

which is the form considered in the Memoir just referred to.

**52.** It is there shown that for an arbitrary position of the satellite line, the parameter  $k$ , or, what is the same thing, the auxiliary parameter  $\theta$ , may be determined by a cubic equation in such manner that the curve shall have a node; the node, or rather the site of the node is termed a *critic centre*; and there are consequently three critic centres (all real or else one real, two imaginary). If however the satellite line touches a certain curve called the envelope, then two of the critic centres unite together, forming a *twofold centre* which is (not a mere node but) a cusp on the corresponding cubic curve; the other critic centre is termed a *one-with-twofold centre*; and the loci of the twofold centre and of the one-with-twofold centre respectively are determinate curves, the former a conic, the latter a cubic. The critic centres corresponding to the entire series of satellite lines which pass through a certain fixed point lie on a cubic, which when the fixed point lies on the line  $x + y + z = 0$ , here the line infinity, degenerates into a conic called the “Harmonic Conic,” and when one of the critic centres is known the other two are determined as the intersections of the harmonic conic by the polar of the given critic centre in regard to the twofold-centre conic.

**53.** For establishing the theory of the groups of the hyperbolas *A*, it is necessary to consider the geometrical forms of the several curves which have been just referred to; viz. this is to be done, on the assumption always that the line  $x + y + z = 0$  is the line infinity, for the Redundant Hyperbolas taking the lines  $x = 0, y = 0, z = 0$  to be real lines and for the Defective Hyperbolas, taking them to be one real and the other two imaginary. The formulae of the

memoir above referred to are in their actual form adapted to the former case, but they can of course be transformed so as to adapt them to the latter case. I proceed to examine the two cases separately.

*The hyperbolas A Redundant (See fig. 1). Article Nos. 54 to 71.*

**54.** Everything is symmetrical with respect to the three asymptotes, and to fix the ideas and without any real loss of generality we may consider the asymptotes as forming an equilateral triangle. Taking the perpendicular distance of a vertex from the opposite side as unity, the absolute magnitudes of the coordinates may be fixed by assuming  $x + y + z = 1$ ;  $x, y, z$  will then denote the perpendicular distances of a point from the three sides respectively. If the coordinates of a point are proportional to  $\alpha, \beta, \gamma$ , then the absolute magnitudes are of course

$$\frac{\alpha}{\alpha + \beta + \gamma}, \quad \frac{\beta}{\alpha + \beta + \gamma}, \quad \frac{\gamma}{\alpha + \beta + \gamma};$$

the point may be spoken of indifferently as the point  $(\alpha, \beta, \gamma)$  or the point

$$\left( \frac{\alpha}{\alpha + \beta + \gamma}, \frac{\beta}{\alpha + \beta + \gamma}, \frac{\gamma}{\alpha + \beta + \gamma} \right),$$

and it is sometimes convenient to use both of the two notations; thus the Harmonic point (that is, the point the harmonic of infinity in regard to the triangle) is the point  $(1, 1, 1)$  or  $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ . The lines  $y - z = 0, z - x = 0, x - y = 0$ , which are the lines joining the harmonic point with the three vertices respectively, are in the case of the equilateral triangle the perpendiculars from the vertices on the three sides respectively, and they may be spoken of simply as the perpendiculars, but it is to be borne in mind that the former is the proper construction of these lines.

**55.** The equation of the envelope is

$$\sqrt{x} + \sqrt{y} + \sqrt{z} = 0,$$

or, what is the same thing,

$$x^4 + y^4 + z^4 - 4(yz^3 + y^3z + zx^3 + z^3x + xy^3 + x^3y) + 6(y^2z^2 + z^2x^2 + x^2y^2) - 124(x^2yz + xy^2z + xyz^2) = 0.$$

The curve consists, as shown in the figure, of a trigonoid branch inscribed in the triangle and of three acnodes outside the triangle.

**56.** The side  $x = 0$  touches the curve in the point  $(0, 1, 1)$  or  $(0, \frac{1}{2}, \frac{1}{2})$ , which is its intersection with the perpendicular  $y - z = 0$ ; the side  $x = 0$  has with the curve at the point in question a four-pointic intersection. The last-mentioned line  $y - z = 0$  meets the curve in the point  $(-4, 1, 1)$  or  $(2, -\frac{1}{2}, -\frac{1}{2})$ , which is one of the acnodes, and therefore a point of twofold intersection; then again in the point  $(16, 1, 1)$  or  $(\frac{8}{9}, \frac{1}{18}, \frac{1}{18})$  which may be considered as a vertex of the trigonoid branch, and finally in the before-mentioned point  $(0, 1, 1)$  or  $(0, \frac{1}{2}, \frac{1}{2})$ , which is the point of contact with the side  $x = 0$ .

**57.** The equation of the twofold-centre locus is

$$\sqrt{x} + \sqrt{y} + \sqrt{z} = 0,$$

or in a rational form

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0,$$

which is, in the case of an equilateral triangle, a circle inscribed in the triangle and touching the sides at their midpoints respectively. The circle is shown in the figure.

**58.** The equation of the one-with-twofold-centre locus is

$$-(-x + y + z)(x - y + z)(x + y - z) + xyz = 0,$$



or, what is the same thing,

$$x^3 + y^3 + z^3 - (yz^2 + y^2z + zx^2 + z^2x + xy^2 + x^2y) + 8xyz = 0.$$

It is a cubic having the harmonic point  $(1, 1, 1)$  or  $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$  for an acnode, touching the sides of the triangle externally at their midpoints respectively, and having the three asymptotes

$$5x - 4y - 4z = 0, \quad -4x + 5y - 4z = 0, \quad -4x - 4y + 5z = 0,$$

or, what is the same thing,  $x = \frac{4}{5}, y = \frac{4}{5}, z = \frac{4}{5}$ ; the form of the curve is shown in the figure.

**59.** The equation of the harmonic conic corresponding to the satellite line  $\lambda x + \mu y + \nu z = 0$  is

$$\frac{\lambda}{x} + \frac{\mu}{y} + \frac{\nu}{z} = 0,$$

or say

$$2fyz + 2gzx + 2hxy = 0,$$

where  $f + g + h = 0$ . It is a hyperbola passing through the angles of the triangle, and through the harmonic point  $(1, 1, 1)$  or  $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ . Observing that the points in question are the intersections of the two rectangular hyperbolas (pairs of lines)  $x(y - z) = 0, y(z - x) = 0$ , it follows that the harmonic conic is a rectangular hyperbola.

**60.** The coordinates of the centre are  $f^2, g^2, h^2$  and the centre is consequently a point on the circle which is the twofold-centre locus.

The asymptotes of course meet in the centre, and they again meet the circle in two points which are the intersections of the circle with the line  $fx + gy + hz = 0$ .

**61.** The Harmonic Conic is the same for the satellite lines which have a given direction, and we may to determine it take a satellite line which touches the envelope. If the constants  $\alpha, \beta, \gamma$  satisfy the condition  $\alpha + \beta + \gamma = 0$ ; then the equation of a satellite line tangent to the envelope is  $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 0$ ; the coordinates of the point of contact with the envelope are as  $\alpha^4 : \beta^4 : \gamma^4$ ; the coordinates of the twofold centre as  $\alpha^2 : \beta^2 : \gamma^2$ ; the coordinates of the one with twofold centre as

$$\alpha^2(\beta - \gamma) : \beta^2(\gamma - \alpha) : \gamma^2(\alpha - \beta).$$

The values of  $f, g, h$  are as  $\alpha^2(\beta^2 - \gamma^2) : \beta^2(\gamma^2 - \alpha^2) : \gamma^2(\alpha^2 - \beta^2)$ , or what is the same thing, as  $\alpha^2(\beta - \gamma) : \beta^2(\gamma - \alpha) : \gamma^2(\alpha - \beta)$ , or as  $\alpha(\beta^2 - \gamma^2) : \beta(\gamma^2 - \alpha^2) : \gamma(\alpha^2 - \beta^2)$ : the last-mentioned values show that the line  $fx + gy + hz = 0$  passes through the harmonic point  $(1, 1, 1)$  or  $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ , and also through the point  $(\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2})$ , the inverse of the point  $(\alpha^2, \beta^2, \gamma^2)$  which is the twofold centre.

**62.** On account of the symmetry of the figure in regard to the three asymptotes, it is sufficient to construct the harmonic conic for a direction of the satellite line inclined to the base at an angle not  $> 30^\circ$ , and this is what is accordingly done: it may however be remarked that for the limiting inclination  $= 0^\circ$ , that is, when the satellite line is parallel to the base, the harmonic conic becomes a pair of right lines, the base and perpendicular; but that for the other limiting inclination  $= 30^\circ$ , that is, when the satellite line is perpendicular to one of the legs of the triangle, the harmonic conic is still a proper hyperbola, and is situate symmetrically in regard to the leg in question; the two limiting cases will be readily understood by means of the general case shown in the figure.

**63.** Imagine now the satellite line moving parallel to itself through the series of positions  $ABCMDEA'$ ; to simplify the figure these are not delineated in their proper positions (but they are merely indicated according to their order of succession), and it is to be understood that they have the following positions, viz.

- $A$ , at infinity;<sup>4</sup>
- $B$ , through the vertex  $B$ ,
- $C$ , touching the envelope,
- $M$ , through the vertex  $M_1$ ,
- $D$ , through the vertex  $D_1$ ,
- $E$ , through node  $X$  of the envelope,
- $A'$ , at infinity;

then the corresponding positions of the critic centres are, on one branch of the hyperbola:  $A_1$  at infinity;  $CM_1$ , a one-with-twofold centre;  $C_1$ , a one-with-twofold centre;  $AE_1$ ;  $A_1$  at infinity. On the other branch:  $A_2$ , at infinity;  $A_3$ , the harmonic point;  $AB_1$ ;  $C_{12}$ , a twofold centre;  $M_1$ ,  $M_2$  are imaginary;  $C'_{12}$ , a twofold centre;  $D_1$ ,  $D_2$ ;  $E_1$ ,  $E_2$ ;  $A'_2$ , at infinity;  $A'_3$ , the harmonic point.

**64.** For the further explanation of the figure it is to be observed that  $B_1$ ,  $B_2$  lie on the line joining the midpoints of two sides; and in like manner  $D_1$ ,  $D_2$  on the line joining the midpoints of two sides; (the imaginary points  $M_1$ ,  $M_2$  are in like manner on the line joining the midpoints of two sides): these relations depend on the theorem, No. 81, of the Memoir on Involution, viz. that for the satellite lines which pass through a vertex  $(1, 0, 0)$  of the triangle, one of the critic centres is the vertex  $(1, 0, 0)$ , and the other two critic centres are points on the line  $-x+y+z=0$ , or, what is the same thing,  $x = \frac{1}{2}$ .

**65.** Again, the point  $E_3$  is on the line  $(x = 1)$  through the vertex  $D$ , parallel to the base, and the points  $E_1$ ,  $E_2$  are on the hyperbola (indicated by a dotted line in the figure)  $(y + \frac{1}{4})(z + \frac{1}{4}) = \frac{1}{16}$ ; this depends on the theorem Nos. 73 and 74 of the Memoir on Involution, viz. the critic centres corresponding to the satellite lines through the point

$$(-4, 1, 1) \text{ or } (2, -\frac{1}{2}, -\frac{1}{2})$$

lie one of them on the line  $y + z = 0$ , and the other two on the conic  $x(x + y + z) - 4yz = 0$ ; reducing by the condition  $x + y + z = 1$ , these equations become respectively  $x = 1$ , and

$$(y + \frac{1}{4})(z + \frac{1}{4}) - \frac{1}{16} = 0.$$

**66.** The foregoing positions of the satellite line, and the critic centres, as exhibited in the figure, were selected partly for facility of delineation; I wished however to examine the effect of the passage of the satellite line through a node of the envelope; and it appears that such passage does not give rise to any marked peculiarity in regard to the critic centres. The selected positions are sufficient to indicate the circumstances of the critic centres as the satellite line passes from the position  $A$  at infinity continuously to the position  $A'$  at infinity; in particular we see that as the line passes from  $A$  to  $C$ , or from  $C'$  to  $A'$ , there are three real centres; but that as the line passes from  $C$  to  $C'$  there is only one real centre.

**67.** The case of the satellite line parallel to the asymptote  $x = 0$ , is included (as already mentioned) as a limiting case in the foregoing one; the harmonic conic is here the pair of lines  $x(y - z) = 0$ : and we have two critic centres on the line  $y - z = 0$ , (the perpendicular), and the third (not properly a critic centre) at infinity on the asymptote  $x = 0$ ; in fact, starting with a critic centre on the line  $y - z = 0$ , the polar of the centre in regard to the twofold-centre conic or circle is a line parallel to the asymptote  $x = 0$ , and which therefore meets the harmonic conic  $x(y - z) = 0$  in a second centre on the line  $y - z = 0$ , and in the point at infinity on the line  $x = 0$ . But the analytical theory of the case is peculiar and may be specially considered.

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<sup>4</sup>Strictly speaking a line at infinity is the line infinity, and as such has no definite direction; but we may of course consider a line which moves parallel to itself in opposite senses as having for its limit the line infinity.

**68.** Writing  $\mu = \nu$ , the equation of the satellite line is  $\lambda x + \mu(y + z) = 0$ , or putting  $x + y + z = 0$  (sic,  $x + y + z = 1$ ) this is  $x = \frac{\mu}{\mu - \lambda}$ . The equation in  $\theta$  (see Memoir on Involution, No. 20) becomes

$$(\theta + \mu)(\theta^2 - \theta\mu - 2\lambda\mu) = 0;$$

or, disregarding the factor  $\theta + \mu = 0$ , which corresponds to the centre at infinity, the equation is

$$\theta^2 - \theta\mu - 2\lambda\mu = 0,$$

which is a quadratic equation, giving therefore two values of  $\theta$ , and the corresponding critic centres lie on the perpendicular  $y - z = 0$ , the  $x$  coordinate being given by the equation

$$\theta = \frac{\lambda x}{\frac{1}{2} - x}.$$

We have therefore conversely

$$\lambda x^2 - \mu x(\frac{1}{2} - x) - 2\mu(\frac{1}{2} - x)^2 = 0,$$

or, what is the same thing,

$$(\lambda - \mu)x^2 + \frac{5}{2}\mu x - \frac{1}{2}\mu = 0,$$

so that putting for shortness  $\frac{\mu}{\mu - \lambda} = \varpi$  (where  $\varpi$  denotes the distance of the satellite line from the asymptote  $x = 0$ ), then the equation which determines the distance of the critic centres from the asymptote is

$$x^2 - \frac{5}{2}\varpi x + \frac{1}{2}\varpi = 0,$$

or we have

$$x = \frac{5}{4}\varpi \pm \frac{1}{4}\sqrt{25\varpi^2 - 8\varpi}.$$

The condition for a twofold centre is ( $\varpi = 0$ , which may be disregarded, or else)  $\varpi = \frac{8}{25}$ , or, what is the same thing,  $8\lambda + \mu = 0$ .

**69.** If  $x_1, x_2$  are the coordinates of the two centres, we have

$$2x_1^2 = \varpi(3x_1 - 1),$$

$$2x_2^2 = \varpi(3x_2 - 1),$$

or, reducing,

$$\frac{x_1}{x_2} = \frac{3x_1 - 1}{3x_2 - 1},$$

$$x_1 + x_2 - 3x_1x_2 = 0,$$

a relation connecting the two values  $x_1, x_2$ ; this equation however only expresses the known relation that the two centres are harmonics of each other in regard to the twofold-centre conic or circle.

**70.** The foregoing examination of the form of the envelope shows very readily what are the positions of the satellite line which give rise to Plücker's groups for the Hyperbolas *A Redundant*. We have in fact first,

*Hyperbolas A Redundant, no osculating asymptote.*

The satellite line is not parallel to a side of the triangle; and the different positions give the following six of Plücker's groups, viz.

I. Satellite line cuts three sides produced.

II. „ passes through a vertex and cuts opposite side produced.

- III. „ passes through a vertex and cuts opposite side.
- IV. „ cuts two sides and a side produced, but does not cut the envelope.
- V. „ touches the envelope.
- VI. „ cuts the envelope.

Next,

*Hyperbolas A Redundant, one osculating asymptote.*

The satellite line is parallel to the osculating asymptote, say to the base of the triangle; and the different positions give the following six of Plücker's groups, viz.

- IX. Satellite line above the vertex.
- X. „ through the vertex.
- XI. „ below the vertex, but not cutting envelope.
- XII. „ touches envelope.
- XIII. „ cuts envelope.
- XIV. „ lies below the base.

And finally,

*Hyperbolas A Redundant, three osculating asymptotes.*

The position of the satellite line is here completely determined, giving one of Plücker's groups, viz.

- XVI. Satellite line at infinity.

**71.** It may be remarked that in this enumeration no account is taken of the nodes of the envelope: the enumeration was in fact made by Plücker by considerations relating to the critic centres, but without arriving at or making use of the envelope at all: if account were taken of the nodes of the envelope several of the foregoing groups would have to be subdivided according to the different positions of the satellite line in regard to these nodes: but the effect produced by the passage of the satellite line through a node of the envelope is so slight, that I am inclined to think that the enumeration may be properly effected in the foregoing manner, without any account being taken of these nodes.

*The Hyperbolas A Defective (See fig. 2). Article Nos. 72 to 101.*

**72.** If in the formulae for the Hyperbolas A Redundant we write

$$\frac{1}{2}(x + yi) \text{ for } x, \quad \frac{1}{2}(x - yi) \text{ for } y, \quad \lambda - \mu i \text{ for } \lambda, \quad \lambda + \mu i \text{ for } \mu,$$

then the equation of the satellite line is

$$\frac{1}{2}(\lambda - \mu i)(x + yi) + \frac{1}{2}(\lambda + \mu i)(x - yi) + \nu z = 0,$$

which is  $\lambda x + \mu y + \nu z = 0$  as before; the equation of the line infinity is  $x + z = 0$ , and the equation of the curve is

$$\frac{1}{4}(x^2 + y^2)z + k(x + z)^2(\lambda x + \mu y + \nu z) = 0.$$

*[The treatment of the Hyperbolas A Defective continues from Art. 73 with the equation of the envelope reduced to a quartic in  $x$  and  $y$ , the discussion of the nodes of the envelope, the twofold-centre conic in the defective case, the one-with-twofold-centre cubic, the harmonic conic for the defective hyperbola, and the analogous enumeration of Plücker's groups XVIII–XXIII for the*

*Defective Hyperbolas A without an osculating asymptote, then groups XXVIII–XXXIII for the cases of a real osculating asymptote, and finally group XXXV for the three osculating asymptotes. The Article Nos. 73–101 carry through this analysis in close parallel with the redundant case treated above, and they continue past the present 50-page extract. The remainder of Paper 350, covering the Hyperbolas B, the Parabolic Hyperbolas, the various Hyperbolisms, the Divergent Parabolas, the Trident Curve, and the Cubical Parabola, falls beyond PDF p. 400 and will be taken up in the subsequent installment.]*

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*End of excerpt: PDF pp. 351–400 of CAYLEY, COLL. MATH. PAPERS, VOL. V.*